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HighP5 Compiler for Hierarchical Distributed Memory Machines

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ABSTRACT

The landscape of high performance parallel computing is currently dominated by distributed, hierarchical machines. Supercomputers and computer clusters now connect multicore processors with local memories through high-speed networks where both the former and the latter are themselves hierarchical in nature due to multi-level caches in NUMA-nodes and complex inter-connection topologies among server racks and cabinets. Careful data and computation placement, low-overhead shared-data synchronization within cores of a node, and optimized communication across nodes over the network are critical for good program performance in present-day machines. For HighP5 programming language, all these aspects are challenging compilation issues as HighP5 is a declarative programming language where programmers write a pseudocode-like parallel program where only intentions regarding these aspects are specified and their proper implementation is left for the compiler to determine. This article discusses how the HighP5 compiler solves the issues related to efficient memory-management, intra-node synchronization, and communication in hierarchical distributed machines and compares performance of HighP5 executables for some building block reference problems against optimized hand-written MPI codes. Although the design decisions and algorithmic elements discussed in this writing are part of the HighP5 compiler; **we believe they are instructive for compiler developers of** any parallel programming language that works in distributed, hierarchical machines and wants to abstract away data synchronization and communication operations from the programmers.

1 | Introduction

The abundance of distributed memory machines in high performance computing is evident from the TOP500 [1] supercomputers list. For several decades running, all the machines in the list are distributed memory machines. In early days, these machines were pure distributed memory machines: connecting uncore CPUs via some high speed networks. However, now almost all supercomputers have multicore CPUs and GPUs interconnected through complex interconnection

networks. For example, the first exascale machine Frontier [2] has one 64-core AMD CPU and 4 integrated GPUs as building block nodes. Consequently, these machines can be called distributed shared-memory multicomputers where parts of the total aggregate distributed memory of the machines are shared among groups of processing units. The same characterization holds for compute clusters prevalent in campuses and organizations, as they comprise commodity machines armed with multicore CPUs and GPUs. This **advance** of distributed shared-memory multicomputers – henceforth, we

will call them just distributed memory machines for simplicity – should continue without some **great change** in the architecture of silicon-based computers due to the inherent scalability problem with hardware cache-coherency protocol [3] for large purely shared-memory computers.

The big question is how to utilize distributed memory machines in high performance computing effectively. Although the extent of exa-scale and peta-scale machines’ may **surprise** an onlooker, actually the available computation power historically always lags behind **the** application demand [4]. The reason is that algorithms for most interesting problems are of quadratic or higher order, while improvement in computation capacity is linear; and larger capacities always drive application interest towards even larger input and higher precision [5]. Since performance is the supreme concern, making the best use of the capacities of underlying execution platform is essential. Exploiting the capacities does not come from just a better parallel algorithms, it requires balancing communication with computation, optimizing data synchronization, and better distribution of computation and data [6] – all of which are platform dependent. Consequently, it is fair to say that high performance parallel computing is hard except for some simple cases.

The desire to **exploit** all features of a target execution platform led to the dominance of low-level primitives or language extensions, such as, MPI [7], Pthreads [8], and CUDA [9] in parallel computing for several decades. However, effective utilization of recent distributed memory machines require combining these primitives that are individually difficult. Furthermore, although the best performance is attainable using these primitives, the primitives themselves do not inherently promote good decision making about a program implementation; rather, programmers’ ad hoc understanding of the machine architecture is the key. For example, it is shown that incorporating network topology-dependent reasoning significantly improves MPI program implementations’ performance [10]. However, with the complexity of both interconnection network and node architecture on the rise, such ad hoc decision making has also become increasingly difficult.

Due to this increasing difficulty of parallel programming in newer machines, the interest of high-level programming languages that abstract away some aspects of data distribution, synchronization, and communication has grown. The approaches in this regard widely varies, from *pragma* pre-processor parallization directives in OpenACC [11] for accelerator-like platforms, multi-paradigm parallel library integrations in Julia [12] for different architectures, detailed data distribution and computation placement facilities in KoKKOS [13], or a simpler cleaner uniform partitioned shared-memory programming in PGAS languages [14].

HighP5 [15] is unique among the new language and library alternatives to low-level coding in the sense that it allows programmers to map computation and data to arbitrary partition levels of a hierarchical distributed memory machine. HighP5 is a declarative programming language where programmers express the flow of computations using a pseudocode-like description and specify a hierarchical data distribution that computations manipulate. They then explicitly ‘map’ the computations and data to different levels of hierarchy of the target execution platform. The compiler

does the necessary index transformation, synchronization insertion, and data communication that are needed for the correct behavior and good performance of the executable code. This breakdown of responsibilities enables efficient executable generation of the same source program for different target platforms using different mapping configurations.

This article discusses the working principle of the HighP5 compiler for distributed memory platforms with a particular emphasis on data part representation, logic, and algorithms for automatic data synchronization and communication. Two critical concerns in this regard are to minimize the cost of decision making code that compiler inserts in-between users’ application logic so that performance debugging is easy and to optimize the implementation of data synchronization and communication logic. The compiler generates MPI+Pthreads hybrid code where synchronization within a shared-memory region is achieved using Pthreads semaphores and barriers and various forms of MPI point-to-point and group communications are employed for syncing data among distributed memory parts. Careful attention is paid to the use of appropriate MPI collective communication primitives whenever possible to take the advantage of separate collective communication networks that are common on modern supercomputers [16] and HPC environments [17][18]. Furthermore, the compiler applies asynchronous communication and code reordering to maximize computation-communication overlap. The article also analyzes the performance of HighP5 executables for three building block problems from dense and sparse linear algebra, and finite-value method in comparison to optimized handwritten benchmark code implemented in plain MPI on the Fugaku supercomputer [19]. The comparative analysis exposes some important implementation aspects of automatic synchronization and communication that greatly impact the performance at scale.

The contribution of this article is important mainly from two respects. First, the techniques and algorithms HighP5 distributed memory compiler uses are not specific to the language. Rather, it treats efficient data-parts representation and communication requirement calculation in a hierarchically partitioned distributed memory machine as generic problems whose solutions should be useful to current and future language compilers that target similar machine architectures. Second, although the literature on languages, platform improvements, and performance assessments is rich in parallel computing; there is a **scarcity** of publications on compilers that this article amplifies.

2 | Related Work

The extent of compilers’ responsibilities in parallelizing high performance computing programs evolved with the evolution of the parallel hardware architectures. We can track early efforts on automatic extraction of fine-grained parallelism from functional languages programs in languages like SISAL [20][21] and VAL [22]. Parallel hardware researchers at that time were focused on building data flow machines [23][24] that can utilize fine-grained parallelism effectively. However, with the lack of success with data-flow machine architectures, interest in fine-grained parallelism extraction also died.

Early successful supercomputers are shared-memory machines that connect a large number of independent processing units to a shared memory [25] [26] [27]. Their success ushered great interest in shared-memory parallel programming, in particular, in parallelizing compilers [28][29]. A parallelizing compiler takes a sequential code, does data flow analyses of the code to identify operations that can be done in parallel, then generates an equivalent parallel program by segregating parallel parts and inserting synchronization operations in appropriate places. However, the scalability problem with concurrent memory accesses from a large number of processors and the difficulty of implementing hardware cache coherence protocol in large machines led to the demise of shared memory machines in high performance computing.

The advent of early distributed memory machines **led to the demise of** parallelizing compilers [30]. The reason behind is the gross disparity between the cost of a communication and the cost of a computation, with former being hundred to thousand times higher than the latter. Foster's influential methodology of parallel programming [6] is a good mechanism to formally explore the underlying problem. According to Foster, efficient parallel programming in a distributed memory machine requires: first, a parallel algorithm; second, determining communication among parallel parts; third, agglomerating the parts into larger units to reduce communications; and fourth, mapping those units to the parallel processors of the target execution platform. Parallelizing compilers were good at the first two tasks. However, generic solution to agglomeration and mapping is a computationally hard problem, thus, unsolvable for compilers.

A key takeaway from that era of high performance computing is that programmers should specify appropriate data distribution and implicitly/explicitly delineate the positions of the code where data synchronization among overlapping data parts located in different places of a distributed memory machine. The compilers should implement the communication and synchronization logic based on programmers' direction. Co-array FORTRAN [31], HPF [32] [33], C** [34], and NESL [35] are examples of languages, among many others, built with this vision at that time. However, the programming model that eventually dominated is the Message Passing Standard (MPI) [36], which is a realization of Hoare's Communicating Sequential Processes (CSP) [37]. MPI is a low-level library programming with C and FORTRAN bindings, where programmers control all aspects of computation and communication, and hardware engineers focus on efficient implementation of different MPI communication primitives. This role breakdown practically relieves the language compiler from all responsibilities related to parallelism.

There are many arguments for MPI's predominance in high performance computing. Giving programmers the utmost control on all aspects of parallel programming undoubtedly promotes the best performance, which is the prime concern, and argued as the main contributors to MPI's success. However, optimized MPI programs are difficult to write and equally difficult to debug. The key problem with the MPI programming is that there is no clean abstraction available to the programmers to reason about the capabilities of the execution platform. Consequently, ad-hoc optimizations are

needed in the code to make the best use of the hardware. Furthermore, a plethora of explicit communication codes hide the algorithmic logic of the implementation.

MPI is still prominent in this era of exascale computing [38]. However, the advent of multicore processors and GPUs in the beginning of this millennium inspires a renewed interest in high-level parallel language alternatives to MPI programming. Supercomputers are built now a days using multicore CPUs and GPUs as nodes connected via one or more high-speed hierarchical interconnection networks. There are regions of memory that are shared among the cores of a node where shared-memory modes of programming, such as, Pthreads [8], and CUDA [39] are appropriate; while, message passing befitting for the distributed network connecting these regions. Consequently, there have been efforts on combining MPIs with shared-memory programming primitives [40] [41][42]. However, such efforts are quite complex and break the MPI model as processes are no longer sequential. In recent years, there have been renewed interest in high-level parallel programming languages as cleaner, productive, and efficient alternatives to low-level parallel programming. The DARPA funded languages such as Chapel [43], X10 [44], Titanium [45], and UPC [46] are prime examples of research and development in this direction. These languages use a partitioned global address space (PGAS) abstraction [14] of the distributed memory machine and allows programmers to specify data distributions in that space. Computations can access both local and remote data where the compiler inserts communication code for copying remote data to the memory of individual local processes based on explicit but typically indirect synchronization instructions. A growing number of supercomputers now include runtime support for these languages along with MPI [47] [48]. With their performance becoming competitive with hybrid MPI code [49], the interest in them should sustain. In fact, some researchers are investigating the possibility of using MPI and threading primitives for runtime engines development for these languages [50].

At the same time, there is **the** realization that new languages and tools should allow utilization of the deep hierarchies at the node and network layers of present day distributed memory machines for best performance from the execution platforms. Hierarchical Tiled Arrays [51], Sequoia [52], Legion [53], and HighP5 [54] are examples of research in this direction. Compilers' responsibilities regarding data synchronization and communication in these languages are more stringent because of the demand for efficient localization of synchronization and communication primitives and due to the tendency of frequent communications at the lower level of the hierarchies in a hierarchical code.

HighP5 compilers are source-to-source compilers. Depending on the execution platform, HighP5 compilers generate low-level threading, MPI, and CUDA codes that are converted into binaries using appropriate downstream compilers. However, a key distinction with some other recent performance portability and multi-platform parallel programming languages such as Kokkos [13], StarPU [55], and Julia [12] is that the HighP5 compilers make all decisions regarding communication, synchronization, GPU offloading, CPU core to task assignments, etc. on their own. The downstream compilers only serves as binary generation tools. In contrast,

Kokkos uses OpenMP [56] and OpenAcc [57] for host-level and accelerator-based parallelization, which are themselves middleware libraries that apply sophisticated hardware related reasoning for efficient executable generation. Similarly, both StarPU and Julia [58] use managed runtime and library-based parallelism for performance portability. In HighP5, language enhancement and compiler enrichment cannot be decoupled due to its unique ‘hardware cognizant’ programming model. This compiler-coupled language development that adheres to the classic principle of supporting only one efficient way of expressing a particular thing in user programs [59] makes HighP5 compilers uniquely interesting.

3 | HighP5 Language Primer

Before we can appreciate the complexity of the HighP5 compiler regarding efficient executable generation, we need to understand how programmer-compiler synergy is achieved in the declarative programming paradigm HighP5 proponents call ‘hardware-cognizant’ parallel programming. For that we first have a brief discussion on the idea of hardware cognizance in programming then use the classical problem of block LU factorization/decomposition from dense linear algebra as a running example to discuss how the notion affects HighP5.

3.1 | Hardware Cognizant Programming

Hardware cognition or awareness in programming is the simple idea that programmers need to design and configure their programs based on the features of the target execution platform to achieve the best performance. This statement is valid even for sequential computing as illustrated in Figure 1 for the matrix-matrix multiplication problem.

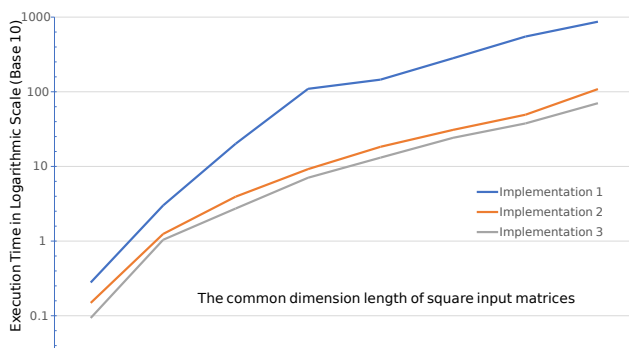


FIGURE 1 | Average execution time (on 10 runs) comparison among three sequential implementations of classic matrix-matrix multiplication program for small square input matrices of size 512^2 to 4096^2 . Here, Implementation 1 is unblocked, Implementation 2 processes matrices in sequence of 256×256 blocks, and Implementation 3 processes them in sequence of 64×64 blocks.

Here, the average execution times of three sequential implementations of $\mathcal{O}(n^3)$ classic matrix-matrix multiplication are

compared for square input matrices of different sizes in an Intel Core i5-1135G7 machine. In the first implementation, no blocking is applied to improve cache-efficiency. That is the implementation is cache-unaware or naive. The second implementation applies matrix blocking but does that without considering the specific cache capacity of the target hardware. Hence, this implementation is cache-conscious. It improves over the first but it is not the best. The third, and the best, implementation configures the matrix blocking size to minimize cache thrashing which requires knowing the cache configuration of the hardware. This is what we understand to be a cache-cognizant implementation. Note that in all three cases the code is compiled with GNU C++ compiler version 11.4.0 with O3 optimization flag enabled. Therefore, despite RAM to cache data movement being controlled by the hardware, programmers still need to specify their ‘intent’ regarding how that should be done by choosing the appropriate block size for dividing the input matrices to attain the best performance. Optimization by the compiler without programmers’ guidance is inadequate.

The dominant paradigms for high-performance parallel computing are hardware-specific (low-level libraries such as MPI, Pthreads, and CUDA) or hardware-agnostic (high-level languages such as Titanium, Julia, Parallel Haskell). Hardware-specific programming enables the utilization of all the salient features of a target hardware. Consequently, they provide the best performance in the relevant architecture, but are not portable and often difficult to use. However, hardware-agnostic programming provides little guidance for the best program design and easily leads to poor performance, although it being portable and easy to use. Due to the critical importance of good performance, high-level languages typically provide platform-specific library integration of low-level building block programs for important/common problems as in Kokkos and Julia as language extensions. However, this approach is not ideal if we want HPC to be learned and used by all. A major concern with high-level parallel programming languages is the performance ambiguity of program constructs in the generated executable for different target execution platforms. This problem is elegantly explained in Snyder’s seminal paper [4] where he shows how the lack of clarity in this regard leads programmer to design sub-optimal parallel programs.

The inherent difficulty of hardware-cognizant parallel programming is that parallel hardware are architecturally quite different and often very complex. Hardware heterogeneity and complexity are the main contributors to the prominence of different low-level tools in the first place. Thus, high-level languages should have difficulty replacing them without compromising performance. The proponents of HighP5 overcome the problem of hardware heterogeneity by introducing a hardware description mechanism (in other words, a type architecture [4]) called PCubeS [60] that can model modern parallel hardware in a uniform way. Figure 2 shows two PCubeS descriptions/models of the Frontier supercomputer [61]. The first model portrays the hardware as a distributed memory computer that exposes all CPU capacities. The second model is for hybrid CPU+GPU computing that is the focus of another HighP5 compiler.

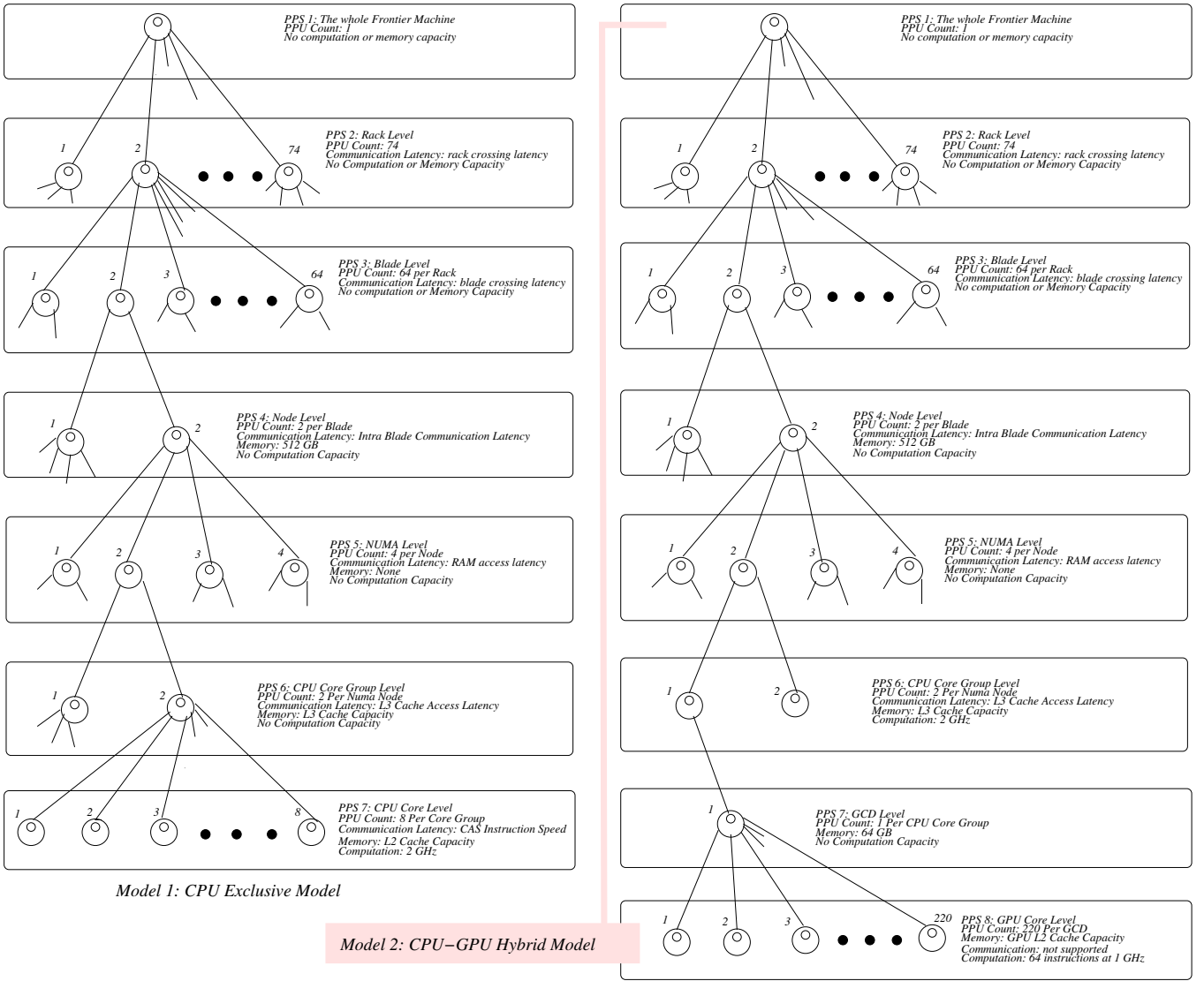


FIGURE 2 | An illustration of two PCubeS machine models of the Frontier Supercomputer. The depictions expand only one parallel processing unit at each level. Furthermore, details of the communication capacities (exact latencies and transaction bandwidth) have not been shown.

PCubeS absorbs hardware-heterogeneity by exposing the architecture-specific features of the execution platform as parts of a symmetric hierarchy of abstract physical processing spaces (PPSes) that are uniformly divided into physical processing units (PPUs) with finite memory, computation, and communication capacities. This exposition of features as capacities enables reasoning about the former without knowing their implementation details and, thereby, facilitates hardware-cognizant parallel programming across hardware platforms. For example, a PPS can be a CPU core, a NUMA node, a single server rack, a GPU, or even an entire region of a computing network. The PPUs of a PPS are the uniform parts of a PPS with similar capacities and communication/interaction delays among them. For example, the CPU cores of a single NUMA node are the PPUs of the core PPS, the individual NUMA nodes of a CPU are the PPUs of the NUMA-node PPS, and the collection of nodes in a single server rack of a supercomputer are the PPUs of the node PPS. All these

hardware elements of different granularities and natures are treated the same.

Hardware-cognizant parallel programming is inherently portable. However, the paradigm requires careful design of the languages that adopt it to ensure that the features of the language have efficient translation to all supported hardware architectures. Otherwise, the same source code will not perform adequately on different execution platforms. At the same time, the paradigm makes platform-specific compiler development challenging as a compiler needs to efficiently translate programmers' design decisions made over an abstract PCubeS hierarchy to appropriate implementations using whatever low-level primitives applicable to the target execution platform. Here, the ultimate goal of the compiler is not to translate an ill-designed source code to an efficient executable – which is impossible – but to generate an executable whose performance will mimic the performance of a low-level implementation had programmers written it using hardware-specific low-level primitives. Given a PCubeS model of the

execution platform has detailed capability specification, programmers are encouraged to incorporate intelligent reasoning about data distribution, computation placement, and execution ordering at different hardware granularity levels in their programs for good performance in the target execution platform. This makes compilation further challenging because the executable's cost and performance should match programmers' expectation. Thus, in contrast to the sequential case, in the parallel case, more sophisticated reasoning ability in the hand of programmers significantly increases the compilation burden.

LU Factorization Algorithms

LU factorization with partial pivoting is one of the most popular methods of decomposing a non-singular square matrix (A) into upper (U) and lower (L) triangular parts. The motivation for the decomposition is to subsequently solve multiple sets of linear equations of the form $A\vec{x} = \vec{b}_i$ using L and U utilizing the property $PA = LU$ hence $LU\vec{x} = P\vec{b}_i$. Here P is the permutation matrix showing how the rows/columns of A have been reordered for triangulation. The basic pseudocode for LU factorization with partial pivoting is shown in Algorithm 1.

Algorithm 1 Basic LU Factorization algorithm

```

1: for  $0 \leq k < n$  do           ▷ Iterate over the rows
2:    $p \leftarrow i \mid \max_{k \leq i \leq n} |A_{ik}|$    ▷ Select the pivot row
3:    $P[k] \leftarrow p$            ▷ Store the pivot row index in an array
4:    $swap(A_k, A_p)$            ▷ Swap pivot with  $k_{th}$  row
5:   for  $k+1 \leq i < n$  do       ▷ Add a new column for L
6:      $L_{ik} = \frac{A_{ik}}{A_{kk}}$ 
7:   end for
8:   for  $k \leq i < n$  do         ▷ Update all rows and
9:     ▷ columns diagonally left and down
10:    for  $k+1 \leq j < n$  do
11:       $A_{ij} = A_{ij} - A_{ik} \times A_{kj}$ 
12:    end for
13:  end for
14: end for
15:  $U \leftarrow A$ 
16: return  $P, L, U$ 

```

Even for a sequential implementation, the crux of the cost of implementing an LU factorization program comes from updating all rows and columns diagonally right and down of the current pivot cell (in Line 8 to 13 of Algorithm 1) that has very poor cache utilization characteristics for large matrices. For improved performance, it is typical to implement an alternative 'blocked' LU factorization that updates only a sliver (that is, a block) of L rows after updating the current pivot column of L . If the block size is b then after b such iterations, the rest of the matrix is updated using a 'saxpy' computation using the updated block of L and U as

$A_{rest} = A_{rest} * L_{block} U_{block}$. An illustration of the block-LU factorization and an algorithm outline are given in Figure 3 and Algorithm 2 respectively.

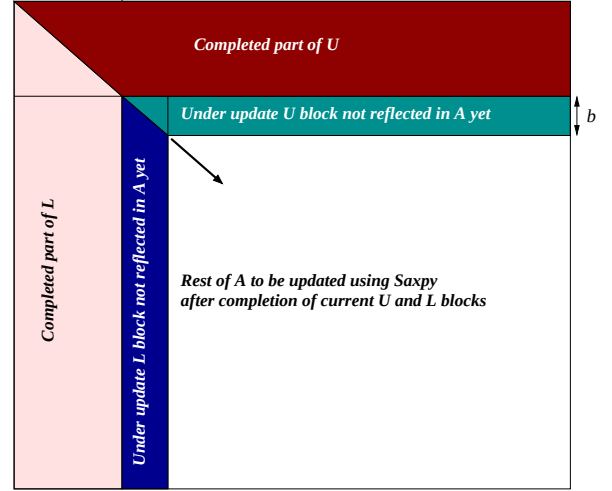


FIGURE 3 | An illustration of the progression of Block LU decomposition strategy.

Algorithm 2 Block-LU Factorization algorithm

```

1: for  $r = 0; r < n; r = r + b$  do       ▷ Iterate rows in blocks
2:   for  $r \leq k < r + b$  do           ▷ Regular LU iterations
3:      $p \leftarrow i \mid \max_{k \leq i \leq n} |A_{ik}|$ 
4:      $P[k] \leftarrow p$ 
5:      $swap(A_k, A_p)$ 
6:     for  $k+1 \leq i < n$  do
7:        $L_{ik} = \frac{A_{ik}}{A_{kk}}$ 
8:     end for
9:     for  $k \leq i < k+b$  do           ▷ Update a sliver of rows
10:      for  $k+1 \leq j < n$  do
11:         $A_{ij} = A_{ij} - A_{ik} \times A_{kj}$ 
12:      end for
13:    end for
14:  end for
15:  for  $r+b \leq i < n$  do           ▷ Saxpy for rest of A
16:    for  $r+b \leq j < n$  do
17:      for  $r+b \leq k < n$  do
18:         $A_{ij} = A_{ij} - A_{ik} \times A_{kj}$ 
19:      end for
20:    end for
21:  end for
22: end for
23:  $U \leftarrow A$ 
24: return  $P, L, U$ 

```

Note that the only difference between Algorithm 1 and Algorithm 2 is of the order of some computation. However, the two algorithms have radically different running times for large A due to divergent cache-usage characteristics when the

saxpy part of Algorithm 2 from Line 15 to 18 is done in small blocks (as in the case of an efficient matrix-matrix multiplication), which Algorithm 1 does not permit. This difference is evidence that high performance computing requires careful consideration of the target architecture.

Parallel LU Factorization and HighP5 Implementation

The parallel algorithm for block-LU factorization is the same as that of Algorithm 2. Only rows/columns of the input and output matrices are distributed among processing units for parallel computations. That is, parallel Block-LU factorization is a data-parallel version of sequential Block-LU factorization. Since LU factorization progresses diagonally downward from the top-left corner to the bottom-right corner of the argument matrix; the norm is to partition the matrix using a block-stride partition where each processing unit takes responsibility for interleaving sequence of blocks of rows/columns so that processing units have roughly equal amount of work to do and are active for most of the duration of the computation, maximizing parallelism.

In HighP5, the computation is expressed as a parallel flow of computation stages happening in a hierarchy of logical processing spaces (LPSes) where each space is partitioned into logical processing units (LPUs) doing data parallel computation of the same stages for different data parts, with support for conditional stage executions. A large program unit such as LU factorization with its own LPS hierarchy is called a ‘Task’ in HighP5 which adheres to the SPMD programming model [62]. Listing 1 shows the parallel computation flow and partition sections of the HighP5 version of parallel block-LU factorization task. The code for the stages and some other sections of the task are not shown, such as, the data structure definition and binding of input/output with task data¹.

```

1 Computation:
2 Space A {
3   Space B { prepareLU(a, u, l) }
4   Repeat for r in a.dimension1.range step block_size {
5     calculateRowRange(a, r, row_range, block_size)
6     Repeat for k in row_range {
7       Space B {
8         Where k in u.local.dimension1.range {
9           selectPivot(Space A: pivot, u, k)
10        }
11      }
12      storePivot(p, k, pivot)
13      Space B {
14        If k != pivot { interchangeRows(pivot, k, u, l) }
15        Where k in l.local.dimension1.range {
16          updateL(u, l, k, l_row)
17        }
18        updateURowsBlock(u, l_row, k, row_range)
19        collectLColParts(l_column, l, k, row_range)
20      }
21      generatePivotColumn(p_column, l_column, row_range, k)
22      Space B {
23        updateUColsBlock(u, p_column, k, row_range)
24      }
25    }
26    Space B {
27      copyUpdatedUBlock(u_block, row_range, u)
28      Where r in l.local.dimension1.range {
29        copyUpdatedLBlock(l_block, row_range, l)
30    }

```

```

31 Space C {
32   Repeat foreach sub-partition {
33     saxpy(u, u_block, l_block, row_range)
34   }
35 }
36 }
37 }
38 }
39 Partition(b, p):
40 Space A <un-partitioned> {
41   a, p, l_column, l_row, p_column, l_block, u_block
42 }
43 Space B <l> divides Space A partitions {
44   a<dim2>, u<dim1>, u_block<dim1>: block_stride(b)
45   l<dim1>, l_column: block_stride(b)
46   l_row, p_column, l_block: replicated
47 }
48 Space C <2t> divides Space B partitions {
49   u: block_size(b, b)
50   u_block: block_size(b), replicated
51   l_block: replicated, block_size(b)
52   Sub-partition <l> <unordere> {
53     u_block<dim2>, l_block<dim1>: block_size(p)
54   }
55 }

```

Listing 1 The parallel computation flow and partitioning section for data distribution of HighP5 block-LU factorization program.

The idea of hardware cognizant parallel programming in HighP5 is that programmers write a declarative program having tasks each depicting a flow of computation stages happening over a task-specific logical hierarchy of LPSes over different data parts in parallel fashion in LPUs of those LPSes. To generate the executable for a specific target platform, programmers map the LPS hierarchy of individual tasks to the various layers of the PPses hierarchy of a PCubeS model of the hardware.

In the Block-LU factorization task of Listing 1, the LPSes are *Space A*, *B* and *C* as we go downward along the LPS hierarchy. The actual degree of parallelism when executing a HighP5 task depends on the nature of data partitioning defined by programmers in the Partition section of the task, how they map LPSes to PPses of the target execution platform, and the number of PPUs in the mapped PPses. The LPUs are multiplexed to the PPUs. Hence, the degree of parallelism increases proportional to the parallel processing capacity of the platform and vary dynamically during a task execution as LPSes of a single task may be mapped to different PPses. Consider two different mappings of Block LU factorization task’s LPSes to the PPses of Frontier architecture’s CPU exclusive model as illustrated below.

```

“Block LU Factorization” {
  Model: CPU Exclusive Model
  Space A: Node Level
  Space B: NMA Level
  Space C: CPU Core Level
}

```

Listing 2 Single node mapping of block-LU factorization

```

“Block LU Factorization” {
  Model: CPU Exclusive Model
  Space A: Rack Level
  Space B: Node Level
  Space C: CPU Core Group Level
}

```

Listing 3 Multi-node mapping of block-LU factorization

¹ The full code is available in the end of [15].

In the first mapping, the un-partitioned root LPS, *Space A* is mapped to a single node. Thus, the entire LU decomposition will be confined to a single node where one CPU from each of the 4 NUMA blocks inside the node will execute the computation stages of *Space B*, such as, *selectPivot*, *updateL*, *updateURowsBlock* of Line 9, 16, and 18 of Listing 1. Finally, 16 cores per NUMA block will participate in the *saxpy* computation (Line 33) for data assigned to the NUMA block. On the other hand, in the second mapping, *Space A* is mapped to the Rack Level, *Space B* to Node Level, and *Space C* to CPU Core Group Level. Therefore, the whole computation will span 64 blades $\times$ 2 nodes = 128 nodes with one CPU per node executing *Space B* computation stages and 8 CPU cores (one for each core group located within a node) collaborating in *saxpy* computation assigned to *Space C*. This is how programmers generate executables of different nature by changing the mapping configuration for the same HighP5 source code. Programmers specify the intent and compiler handles the implementation, as stated earlier.

The separation of mapping enables portability across architectures also, where different platform-specific HighP5 compilers generate mapping-appropriate executable. However, for our present discussion, we focus on generating different executables for the distributed memory architecture exclusively. We close this section with a discussion on the nature of a HighP5 executable for distributed memory architectures to understand the general principle of executable generation and consequent challenges the compiler faces regarding properly implementing programmers' decisions.

3.2 | The Nature of a HighP5 Executable

HighP5 allows programmers to control and tune all aspects of parallel program design based on the knowledge of the hardware's PCubeS models. However, to keep programmers' job manageable, the programming model is made *declarative*. Programmers specify which computation stages should execute where in hardware without worrying how the loops and statements are translated. They define what data parts to be available where through the partition specification without knowing the complexities of memory management. Finally, they designate the points where shared/overlapped data parts should be synchronized by setting the LPS transition boundaries in a task's Computation Section without dealing with the nitty-gritty of message passing and synchronization primitives. Given the abstract machine model of the execution platform has detailed capability specification, the cost of all these decisions is clear to the programmers after the mapping of LPS hierarchies of various tasks to the PPS hierarchy of the execution platform during program compilation.

To keep the performance cost assessment true to programmers expectation, the overhead code that compiler adds and the memory footprint of management data structures must be negligible, the implementation of automatic data synchronizations must be optimal, and the compiler should not add any hidden communication/synchronization that the most efficient low-level code could avoid.

The HighP5 compiler tackles these challenges by generating an optimized C code from HighP5 program that is parallelized using MPI and POSIX threads (Pthreads). This C code is then compiled into the final executable using some nested MPI-C compiler optimum for the target platform. Consequently, the executable for a HighP5 program is an MPI application where each MPI process itself is parallelized using multi-threading.

Any data synchronization in the source HighP5 program that is confined within a shared memory segment of the execution platform is implemented using Pthreads synchronization primitives such as semaphores and barriers. Consequently, no communication is involved. When a data synchronization spans across shared memory sub-regions of the distributed memory machine, an appropriate message passing primitive is used from the MPI library.

The MPI+Pthread hybrid is chosen for several reasons. At the process-end, MPI is ubiquitous in compute clusters and supercomputers. MPI communication primitives have efficient implementations in most, if not all, supercomputers. Furthermore, MPI supports sophisticated grouping of processes using custom communicators that enable restricting communications within particular segments of a distributed network. Finally, MPI communication is sufficiently low-level and configurable for reliable assessment of various aspects of their costs. All these aspects make MPI an ideal choice for cross-node interactions in a HighP5 executable.

At the threading-end, Pthreads allows pinning threads to specific CPU cores. Thread priority and maximum stack size are configurable in Pthreads. Furthermore, Pthreads use low-level synchronization primitives, enabling custom-code to be plugged in-between synchronization steps. In fact, due to all these features, often other high-level threading alternatives such as GNU C implementation of OpenMP and Apple's grand central dispatch (GCD) [63] are implemented using Pthreads. Hence, Pthreads is ideal for intra-node CPU/CPU-core management in a HighP5 executable.

Since the interim code is a MPI+Pthreads hybrid, the execution hierarchy has only two levels: MPI processes at upper level hosting POSIX threads under them. The compiler decides about the number of processes and threads based on the LPS-PPS mappings of the various tasks of a HighP5 program. Consider the mapping of Listing 3 of the block-LU factorization program to Frontier machine. Since shared memory segments are the nodes of the machine, the number of MPI processes for the HighP5 program will be 128, the total number of nodes within a rack. The number of Pthreads within each MPI process will be 8, the number of CPU core groups within a node. Here each Pthread will be pinned to the first core of a core group.

The idea is that each MPI process and Pthread works as a composite PPU controller and executes code and engages in synchronization and communication depending on its role assignments. For example, in block-LU factorization, the top-most LPS, *Space A*, is non-partitioned. Consequently, only the first Pthread of the first MPI process will execute the computation stages assigned to *Space A*. All Pthreads in all MPI processes will execute computation stages for *Space C* for LPUs multiplexed to them. Meanwhile, only the thread at core i with $i = 0 \pmod{8}$ of an MPI process will execute

computation stages assigned to *Space B* LPUs. The LPUs are evenly multiplexed to the PPU controllers that have roles in the underlying LPS.

When there are multiple tasks in the source program, the union of all task's processing requirements and the capacity available in the hardware determine the number of processes and threads. If capacity is restricted, a single MPI process may participate in two different tasks in a lockstep fashion. On the other hand, when more capacity is available, tasks execute concurrently in different MPI processes if there is no inter-task data dependency. In all cases, a single part of the machine is never assigned to multiple tasks simultaneously. The above strategy ensures that runtime parallelism scales proportionally to the parallel processing capacity of the hardware while, at the same time, the overhead of process and thread management is minimal.

In case of multi-task programs, HighP5 offers the notion of tasks executing in an environment consisting of data parts distributed among MPI processes where execution of a task modifies parts of the environment. The MPI processes last for the entire runtime of the program, the Pthreads inside the processes are cleaned up and spawned again based on the nature of individual tasks' LPS hierarchy. When there are data dependencies among tasks, a subsequent task gets the up-to-date data left in the environment by earlier tasks. Therefore, there is no accumulation and re-distribution of data during transitions between tasks unless the dependent task's data distribution specification is different from how the data is currently residing in the collective MPI environment. Figure 4 illustrates the runtime model of an HighP5 executable. Here, the garbage collector kicks in at the end of the execution of each task to cleanup management data structures and intermediate outputs of the task not needed any further.

Individual composite PPU controller threads for a task are basically flow executors that traverse a flow template and execute a computation stage or skip it depending on its LPS role assignments. For each computation stage, the data parts for the underlying LPU come from the shared data parts registry that is managed by the task executor. As a flow executor reaches a point that requires a synchronization/communication, it retrieves relevant primitives from the data dependency resolver registry and invokes the appropriate sync function. The execution model of such a thread is illustrated in Figure 5.

Finally, note that the programming paradigm encourages programmers to choose data distribution and computation assignment based on their understanding of the PCubeS description of the target platform that exposes all layers of the interconnection network in the machine and cache-organization inside individual nodes. Consequently, implementing data movements from RAM to caches as programmer-intended and confining communications within only relevant nodes are part of compiler's responsibilities. All these along with a careful translation of computation stages and synchronization/communication directives from the HighP5 program to the hybrid MPI+Pthread intermediate is essential to exhibit a performance similar to a hand-written optimized low-level code.

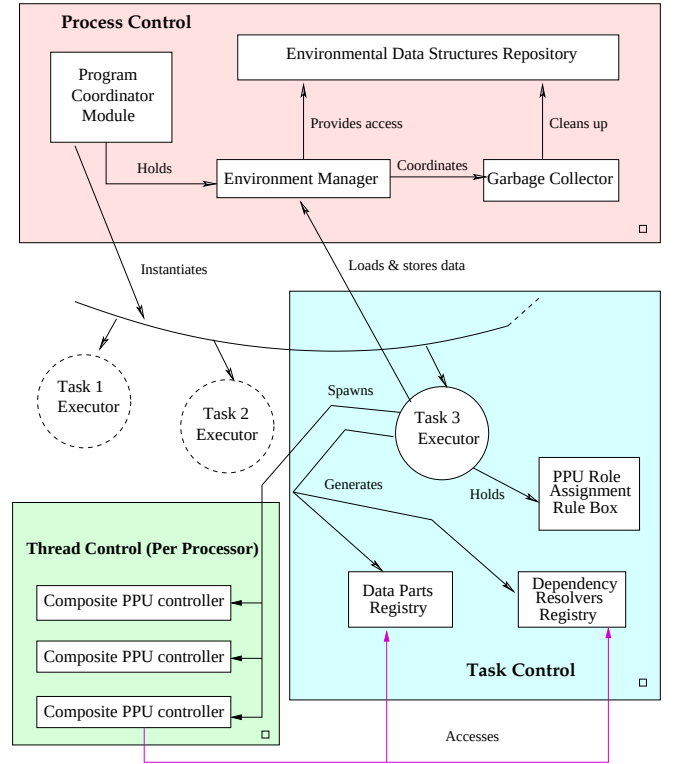


FIGURE 4 | The runtime process model of HighP5 executables.

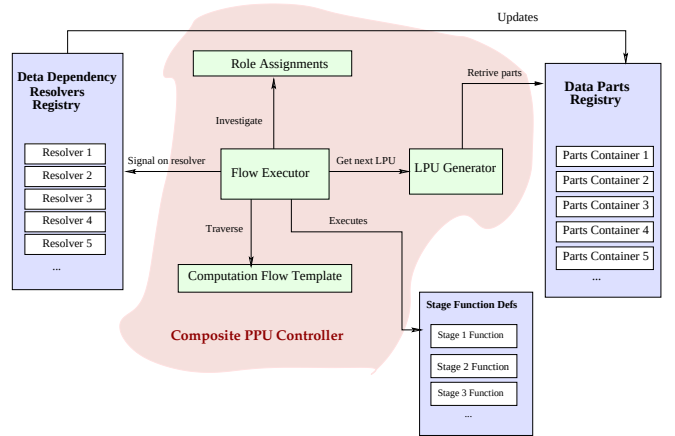


FIGURE 5 | The architecture of a composite PPU controller thread.

4 | Overview of the Compiler

HighP5 offers a paradigm of parallel programming in which the same source code is portable to different architectural platforms. Programmers write the source code using abstract LPS hierarchies to have a flexible declarative program that can be mapped appropriately to platform-specific features to derive an efficient executable for the target platform. To enable this feat, the HighP5 compiler has the breakdown where the compiler front-end doing lexical, syntax, semantic, and static analyses generates an intermediate form using the source code exclusively, ignoring the programmer-supplied

mapping configuration. The back-end compiler has the understanding of architecture-dependent interpretation of PPS hierarchy of the platform. The mapping configuration is fed to this back-end for further analyses before code generation can begin. For the distributed-memory HighP5 compiler this responsibility breakdown is illustrated in Figure 6.

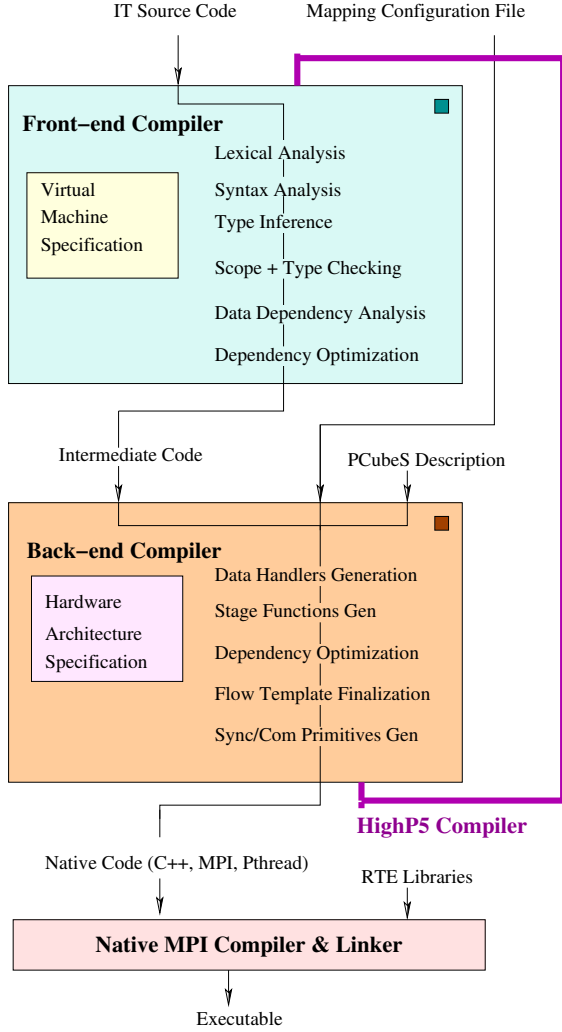


FIGURE 6 | Front-end/Back-end breakdown of the compiler.

HighP5 is a partially and strongly typed language. Only the types of ‘task global’ variables that are shared across processes and stay alive for the duration of a task execution are defined in a specific ‘Define’ section of a task and the types of all remaining local and global variables (including I/O) are inferred. Consequently, there is a type inference step after syntax analysis that kicks off the semantic analysis phase. The concept of partial typing is illustrated in Listing 4 showing the Define and key parts of the Stages section of block-LU factorization task. The type inference is done using a fixed-point calculation. That traverses the abstract syntax tree of the whole program and infers types of new variables and expressions from that of those whose types are already known.

The traversal is repeated as long as new types are discovered or it ends with no further type discovery.

```

1 Define:
2   a, u, l: 2d Array of Real double-precision
3   p: 1d Array of Integer
4   l_row, l_column, p_column: 1d Array of Real double-precision
5   u_block, l_block: 2d Array of Real double-precision
6   pivot: Integer Reduction
7   k, r, block_size: Integer
8   row_range: Range
9 Stages:
10  selectPivot(pivot, u, k) {
11    do { reduce(pivot, "max_entry", u[k][j])
12      } for j in u and j >= k
13  }
14  storePivot(p, k, pivot) {
15    p[k] = pivot
16  }
17  interchangeRows(pivot, k, u, l) {
18    do { pivot_entry = u[i][k]
19      } for i in u and i >= k
20    u[i][k] = u[i][pivot]
21    u[i][pivot] = pivot_entry
22  } for i in l and i < k
23  do { pivot_entry = l[i][k]
24    l[i][k] = l[i][pivot]
25    l[i][pivot] = pivot_entry
26  } for i in l and i < k
27  updateL(u, l, k, l_row) {
28    do { l[k][j] = u[k][j] / u[k][k]
29      } for j in l and j > k
30    u[k][j] = 0
31    l_row[j] = l[k][j]
32  } for j in l and j > k
33  updateURowsBlock(u, l_row, k, row_range) {
34    do { u[i][j] = u[i][j] - l_row[j] * u[i][k]
35      } for i, j in u and i > k
36    and i <= row_range.max and j > k
37  }
38  collectLColParts(l_column, l, k, row_range) {
39    do { l_column[i] = l[i][k]
40      } for i in l and i >= row_range.min and i < k
41  }
42  generatePivotColumn(p_column, l_column, row_range, k) {
43    do { p_column[i] = l_column[i]
44      } for i in l_column and i >= row_range.min and i < k
45  }
46  saxpy(u, u_block, l_block, row_range) {
47    do {
48      total = 0.000000
49      do { total = total + u_block[i][m] * l_block[m][j]
50        } for m in u_block
51      u[i][j] = u[i][j] - total
52    } for i, j in u
53    and i > row_range.max and j > row_range.max
54  }

```

Listing 4 Define and a part of Stages sections of the block-LU factorization task.

If type inference is successful, then scope + type checking starts, which is no different from how this step is done in other languages. However, afterward, the compiler needs to invoke sophisticated static analysis to determine the optimum read-write dependencies among compute stages for shared data. Note that the front-end compiler cannot determine how to resolve those dependencies as it operates using the notion of logical LPS hierarchies of tasks. However, it can eliminate many redundant dependencies based on tasks’ partition specifications and placement of stage functions in LPSes. The final output of the front-end compiler is a collection of abstract flow templates for individual tasks, the abstract syntax tree for the coordinator program that launches tasks, intra

and inter-task data dependency graphs, and symbol tables for various scopes.

The back-end compiler understands how to deal with shared memory parts of individual nodes of the distributed memory machines, the logic of pinning threads to processing cores for better cache utilization, and the logic of pair-to-pair and group communication in a generic way. Configured with the PCubeS description of the specific target execution platform and supplied with programmers' mapping configuration file, it is able to translate the generic rules to specific C code parallelized with MPI and Pthreads. The execution mechanism of composite PPU controller threads as shown in Figure 5 is uniform for all tasks for a specific type of execution platform architecture and, henceforth, comes from a native format runtime environment (RTE) library. The back-end compiler applies complex multi-step analyses to populate the data parts and data dependency resolver registries, configure roles for PPU controller threads and MPI processes, and generate C functions for computation stages and the flow templates. Similarly, the environment management mechanism and garbage collection logic of Figure 4 are parts of the common runtime. The back-end compiler generates the program coordination function (works as the main function of the final MPI program), task executor functions for individual tasks (handle initialization of threads and registries), and environment instructions that link inter-dependent tasks and I/O operations. The advantage of adopting environment instructions for linking is that individual task compilations can be treated as independent of one another that made code optimization easier.

The key concerns related to efficiency of the generated executable is as follows:

1. **Task Data Management:** involves efficient data handlers generation for Data Parts Registry and data access optimization.
2. **Flow Execution Code Generation:** involves appropriate code generation for stage functions and cache-aware multiplexing and iteration of LPUs.
3. **Synchronization and Communication:** involves data dependency optimization, proper sync/communication primitive generation, and maximizing computation-computation overlaps.
4. **Environment Management:** involves generating proper environment instructions and handler code.

Among the above, the first three are dealt with on individual tasks basis and the last by considering inter-task dependencies. We cover these topics in detail in the following sections.

5 | Task Data Management

Key aspects of managing the data relevant to a task is to interpret and translate the hierarchical data distribution specifications found in task's Partition section, generating code for allocating data parts and LPU metadata inside MPI processes and populating the parts, and accessing the parts during stage executions over LPUs by composite PPU controller threads. We discuss these concerns in the following subsections.

Note that these aspects of data distribution management and data access are orthogonal to the target implementation choice being MPI+Pthread hybrid. Rather, they are mathematical problems that compilers of any language that exposes the hierarchical memory organization common in modern architectures (e.g., Sequoia [52] and Legion [53]) need to solve. The efficiency of these problem solutions strongly affects an executable's runtime. However, there is an often-overlooked and important concern of memory scalability of parallel programs. Even for typical parallel algorithms, space complexity per-process reduces at a lower rate than time complexity. Hence, how large program input can be for a fixed number of processors is often limited by the per-processor memory requirements. Therefore, the compiler must ensure that there is no hidden memory overhead that causes the executable's space consumption characteristics to noticeably deviate from the programmers' expectation.

5.1 | Handling Hierarchical Data Distributions

HighP5 supports four standard data distribution specifications for partitioning an array dimension: block-size, block-count, stride, and block-stride. Let us investigate their behavior on an array dimension of length N .

1. **block-size(n)** divides N into $\frac{N+(n*1)}{n}$ blocks each owning n consecutive indices along the dimension, except for the last block which can be smaller.
2. **block-count(c)** divides N into c consecutive index blocks each of size $\frac{N+(c*1)}{c}$. Again, the last block can be smaller if c does not divide N .
3. **stride(n)** divides N into n partitions where the i^{th} partition owns the indices $i, n+i, 2n+i, 3n+i, \dots$.
4. **block-stride(b, n)** divides N into n partitions where the i^{th} partition owns the indices $(bi, bi+b*1), (bni, bni+b*1), (2bni, 2bni+b*1), \dots$. Here (a, b) means the inclusive range from a to b .

For a multidimensional array, different dimensions can be partitioned using different data distribution schemes in each LPS. These four data distributions are well known from the days of FORTRAN D [64] and cover most parallel programming problems. For irregular data structures, such as a graph, the recommendation is to use their array representations (e.g., adjacency matrix, CSR, COO) and partition the array(s). Finally, programmers can left some array dimensions un-partitioned in specific LPS or replicate content along some dimensions.

Having the data distributions of the FORTRAN D era only as the building block partitioning functions may seem limiting. However, given the data distribution is hierarchical and multidimensional, these building blocks can be combined in interesting ways to cover many complex cases. Second, unless a data distribution is always specified as an interval function, its compact representation is generally difficult to construct, resulting in a significant increase of hidden memory overhead. Third, all current technologies for data transfer in networks (e.g. MPI and GASNet [65]), among host CPU and attached accelerators (e.g. PCIe and NVLink), and between RAM and caches operate in block transfer mode. Hence, point-wise data

distributions do not have efficient implementations in current architectures in general. Nevertheless, we admit that there are benefits in supporting non-trivial data distribution functions in HPC languages, in particular, for the sake of programming simplicity. However, HighP5's vision is to incorporate them in the future as libraries – instead of – as language components to adhere to the classic language philosophy of having a small set of core abstractions having efficient translations across execution platforms [59].

Note that block-size and stride distributions are special cases of the block-stride distribution function. In all four cases, we can describe the indices of a dimension assigned to a specific LPU by the partitioning scheme as an interval function of the form $f(\beta, \eta, \rho, \zeta)$. Here β is the starting index, η is the ending index, ρ is the periodic interval, and ζ is the index range within a periodic interval that belongs to the LPU under concern. Given partitioning is hierarchical, the interval function of the LPU divides the index range assigned to its ancestor LPUs in upper LPSes. If we order the LPSes that partition an array dimension from lower to upper positions in the hierarchy as $l_1, l_2, l_3, \dots, l_L$ and use the notation f_{l_i} to designate the data distribution function at l_i then the hierarchical interval range function δ_{l_1} for the lowest LPU can be expressed as a function composition as follows:

$$\delta_{l_1} = f_{l_1} \circ f_{l_2} \circ f_{l_3} \cdots \circ f_{l_L} \quad (1)$$

Figure 7 illustrates the composition of data distribution using a two layer hierarchy where the upper LPS uses block-stride partitioning and the lower LPS uses block-count partitioning functions.

For a D dimensional array, the index ranges allocated to the LPU for all dimensions can be expressed as the sum of Equation 1 as follows:

$$\Delta_{l_1} = \sum_{d=1}^D f_{l_1}^d \circ f_{l_2}^d \circ f_{l_3}^d \cdots \circ f_{l_L}^d \quad (2)$$

Here, the superscript for the distribution functions indicates that the functions can be different for different dimensions. Note that values for the configuration parameters for the component functions of a hierarchical distribution configuration are available mostly at runtime. Therefore, the compiler generates hierarchical data structures that are initialized using command-line parameters, execution-time configuration (e.g., MPI process count), and metadata from input files (e.g., dimension lengths of individual arrays) during the launch of a task executor (Figure 4).

5.2 | Memory Allocation in MPI Processes

HighP5 is an array-based language in the sense that the prime focus of data distribution is arrays. Buffer allocation for lists and other structures is simple as they are replicated across LPUs of an LPS. Consequently, a single whole copy can be used for all LPUs' computation stages inside an MPI process. Buffer allocation for array parts and loading (from input files) and retrieving (from remote memory locations) array items in those buffers requires ingenuity.

If $f(\beta_i^j, \eta_i, \rho_i, \zeta_i)$ is the interval function for a particular LPU l_j for partitioning a dimension i of an array A , the total number

of index points along that dimension, l_i^j , for l_j follows the equation:

$$l_i^j = \zeta_i \frac{\eta_i * \beta_i + 1}{\rho_i} \quad (3)$$

If A is D dimensional then the buffer capacity requirement for the part of A assigned to l_j is:

$$I_j = \text{size}(A.\text{elementType}) \prod_{i=1}^D l_i^j \quad (4)$$

As data distribution is hierarchical, the index points calculation for a lower level LPU l_j can be done by repeatedly applying Equation 3 from topmost ancestor LPU l_1 down to l_j by treating l_k as η_{k+1} for all $1 \leq k < j$. The overall buffer capacity requirement for a single array's parts for all LPUs of an LPS multiplexed to an MPI process can be computed by summing up Equation 4 for all LPUs. The total capacity requirements calculation can be optimized by utilizing the fact that consecutive LPUs are multiplexed in a single PPU controller process. Consequently, LPUs' data distribution interval functions can be combined as follows:

$$\sum_{j=1}^{j=n} f(\beta_i^j, \eta_i, \rho_i, \zeta_i) = f(\beta_i^1, \eta_i, \rho_i, n\zeta_i) \quad (5)$$

The compiler generated intermediate C code utilizes this last equation to determine the buffer capacity requirement for each array inside an MPI process in constant time and allocates the buffer. Allocating all the parts of an array needed inside an MPI process as a single contiguous buffer has the advantage that when garbage collection takes place, the entire buffer can be freed in a single step, eliminating all associated data parts. **Furthermore, if there are overlappings in different LPUs' data parts then overlapped regions are not replicated, which is critical for memory efficiency.**

Note that programmers may define an LPS hierarchy where two different LPSes from different branches of the hierarchy requiring the same array for their computation stages but using different data distribution strategies. If some LPUs of both these LPSes are multiplexed to a single MPI process then there is no feasible way to maintain a single buffer for all data items the process needs to hold. Furthermore, such an attempt reduces the chance of parallel computation. Therefore, an MPI process maintains a separate buffer for each distinct maximal path from the root of the partition hierarchy to the lowest that requires the underlying array. Buffer synchronization within a process for a pair of LPUs not related via an ancestor-descendant relationship is treated as a communication problem.

An important aspect of a path in the LPS hierarchy is that each LPS along the path is dimensional. Consequently, the LPUs can be hierarchically ordered based on the number of partitions in the LPSes and the position of the LPU in the hierarchical order, which gives us a hierarchical LPU ID for each LPU. An array part that is needed by a particular LPU can be given a hierarchical multi-dimensional part-ID derived from the LPU ID. The compiler utilizes this fact to generate code for a part-ID-tree construction for individual buffers. The leaves of the tree refer/point to unique data parts, each pointing to the beginning of a continuous stretch of indices in the buffer. That is, the elements of a single data part are

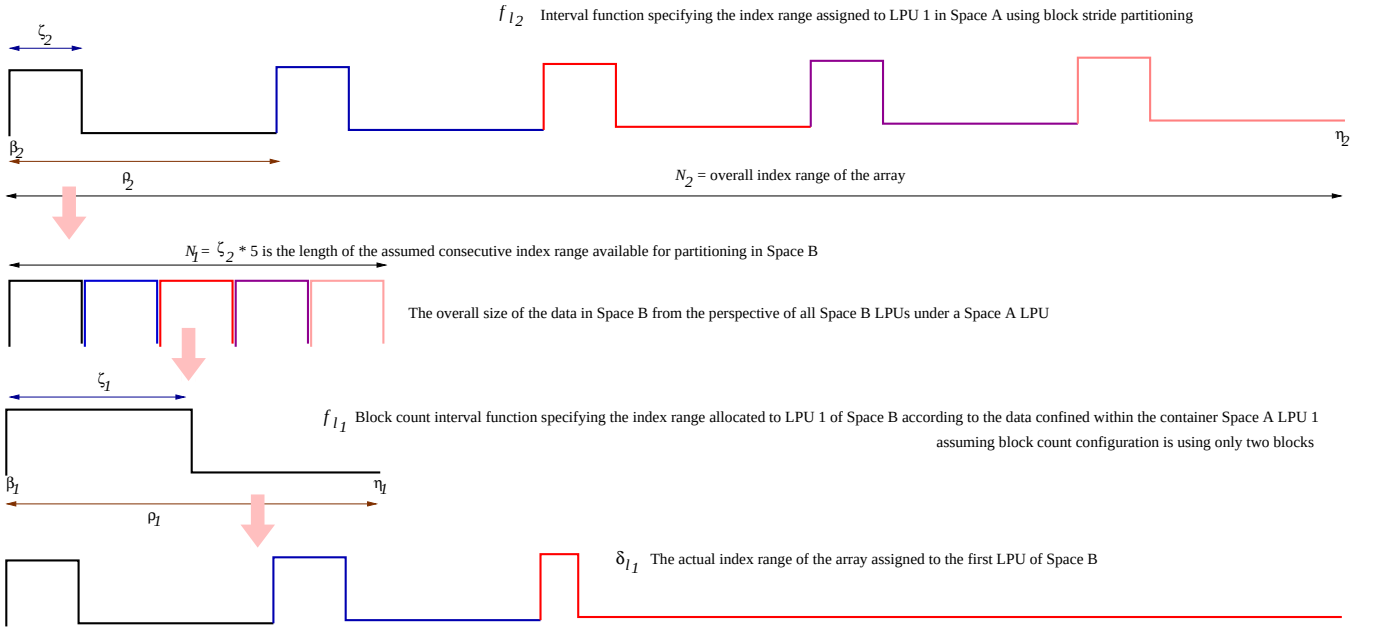


FIGURE 7 | An illustration of nested data partition using a two layer hierarchy where Space A is the upper and Space B is the lower LPS. The lower layer sees a compacted index space created by the upper layer's blocks.

always located in consecutive indices regardless of how the partition configuration may have re-ordered the indices of the original array. This treatment is important to ensure that the data for an LPU computation can fit in the memory of the PPU programmers used for the mapping and that there is no opaque data access overhead that cannot be easily debugged. The portion of the Data Part Registry code (Figure 5) that belongs to the RTE library is extended by compiler-generated code for buffer allocation, part-ID-tree construction, and parts retrieval by IDs for each array and LPS combination. Collectively, they form a data parts container and the registry holds a map of such containers. For the first HighP5 task that needs an array A for its computation, data items are read into the allocated buffer from input files concurrently in all MPI processes collectively by PPU controller threads as part of the registry initialization process. Initialization of the registry for any subsequent HighP5 tasks that require A , data item loading involves cross-task communications, which we discuss later.

5.2.1 | LPU Metadata Management

since HighP5's hierarchical machine model exposes all physical processing spaces (PPS) and their units (PPU), programmers are encouraged to choose partition parameters that promote best cache utilization, optimize data transfer cost, and maximize parallel execution. Consequently, the number of LPUs can be quite large in the lower LPSes. To give an example, if the block-size $b = 64$ and the argument matrix is 10240×10240 in the block-LU factorization problem of Listing 1, then the number of LPUs in Space C will be $(\frac{10240}{64})^2 = 25600$ and the number of LPUs for each sub-partition of Space C LPU will be $\frac{10240}{64} = 160$. Therefore, a straightforward implementation of MPI processes maintaining a list of LPUs for each LPS they are responsible for and

executing computing stages in Pthreads by traversing the list may have a disastrous effect. The LPU tracking lists themselves can overwhelm the memory capacities of the computing nodes. The clever work around this problem is to port translated partition specifications of a task of the source program directly into the intermediate C code.

The compiler generates stateful classes for generating LPU metadata for various LPSes whose instances are created with runtime partition configuration and metadata related to relevant data structures for each composite PPU controller thread. A composite PPU controller thread has multiple static IDs (or ranks) for various PPSes based on the exact hardware units of the execution platform's PCubeS model they represent. As a thread enters/re-enters a new LPS in the flow execution template of the task, it initializes/resets an LPU iterator in the LPU generator with its associated PPU IDs. Here, the PPU IDs are used by the generator class to determine which LPUs are being multiplexed locally and assigned to the threads. The LPU generator computes and returns the next LPU each time the thread asks for one until the LPUs of the thread are exhausted.

An LPU generation involves calculating an LPU ID, deriving various data part IDs from the LPU ID, setting up data part pointers to allocated buffers, and generating part metadata (e.g., a data part size and dimension ranges). It may appear that computing and recomputing LPUs in this manner is costly. However, compared to the execution cost of functions for computation stages, the LPU generation cost is typically insignificant. Furthermore, all calculations related to LPU generation are integer computations that are further inlined in a single function by the final MPI-C compiler. Thus, this strategy effectively eliminates memory footprint for LPU metadata management in exchange for some integer computation overhead.

5.3 | Data Access During LPU Computation

The logic for accessing data parts while executing a computation stage on a particular LPU is rather simple. Since LPU metadata has data part IDs for each array part, it is possible to search the associated part-id-tree in the Data Parts Registry to retrieve the location pointer to the buffer for that data part using the hierarchical part ID. This is done during the generation of the LPU. Subsequent read-write on the array as part of the execution of the computation stages is done using that location pointer. Figure 8 shows how the part retrieval process works for Space C LPUs of the block-LU factorization code of Listing 1 during the execution of ‘saxpy’ computation stage.

For task-global scalars and list type variables, a single global data structure instance is generated by the compiler that holds their memory references. All composite PPU controller threads access these variables through that single instance.

6 | Flow Execution Code Generation

From a high-level perspective, the flow execution code generation is the activity of merging the compute stages of the Stage section of a task (e.g., Listing 4) to its Computation section (e.g., Listing 1) to have a pthreads ‘run’ function that every composite PPU controller thread executes. That different threads will operate on different data is ensured by LPU multiplexing based on PPU IDs. Since HighP5 stage functions are type polymorphic, the compiler often generates multiple type-specific C++ implementations for a single stage function based on its calling context. The parameters for the C++ functions are modified to only access the current LPU data structure, task specific metadata, and PPU controller specific information. This modification is essential to simply synchronization and communication code generation which we discuss later. Listing 5 shows a part of the generated thread run function of the block-LU factorization task that invokes the ‘selectPivot’ stage as an illustration of the stage execution logic.

```
1 { // scope entrance for iterating LPUs of Space B
2   int spaceBLpuId = INVALID_ID;
3   int spaceBIteration = 0;
4   SpaceB_LPU *spaceBLpu = NULL;
5   LPU *lpu = NULL;
6   while((lpu = threadState->getNextLpu(Space_B,
7     Space_A, spaceBLpuId)) != NULL) {
8     spaceBLpu = (SpaceB_LPU*) lpu;
9     { // scope entrance for conditional subflow
10      Dimension uPartDims[2];
11      Dimension uStoreDims[2];
12      uPartDims[0] = spaceBLpu->uPartDims[0].partition;
13      uStoreDims[0] = spaceBLpu->uPartDims[0].storage;
14      uPartDims[1] = spaceBLpu->uPartDims[1].partition;
15      uStoreDims[1] = spaceBLpu->uPartDims[1].storage;
16      if((xformIndex = threadLocals->k,
17        partConfig = *(&spaceBLpu->uPartDims[0]),
18        (xformIndex % (partition.b * partConfig.count))
19        / partition.b == partConfig.index)) {
20        if (threadState->isValidPpu(Space_B)) {
21          // invoking user computation
22          int stage10Executed = selectpivot_stage_9(spaceBLpu,
23            arrayMetadata,
24            taskGlobals,
25            threadLocals,
26            reductionResultsMap,
```

```
27            partition,
28            threadState->threadLog);
29        }
30      } // end of condition checking block
31    } // scope exit for conditional subflow
32    spaceBLpuId = spaceBLpu->id;
33    spaceBIteration++;
34  }
35  threadState->removeIterationBound(Space_A);
36} // scope exit for iterating LPUs of Space B
```

Listing 5 A small portion of the thread run function for block-LUF showing how stage functions are invoked for LPUs in interim C++ code.

There are several subtle performance issues that make code generation for a task’s flow executor a more involved process than just invoking translated C++ stage functions on multiplexed LPUs.

First, given programmers are encouraged to fully utilize the various cache layers exposed by the PCubeS description of a target platform, the LPU hierarchy can be quite deep in a task, with small data parts inside the lower level LPUs. Consequently, the ratio between the amount of calculations as part of a stage computation and that for generating the LPU metadata can be quite low, giving us a large computation overhead. The compiler applies several optimizations to mitigate this problem. First, LPU metadata generator uses the same LPU object and only updates its relevant ID parts in the LPU iteration process described earlier as the PPU controller proceeds from one LPU to the next. Second, iterators are maintained over each associated part-ID-tree also to avoid traversing the trees from the root every time only a small part of the LPU ID changes. In fact, quite often some data part references remain the same as computation flows from LPUs to LPUs. Finally, LPUs are multiplexed to PPU controllers as continuous swaths to minimize up-down movements along part-id-trees as the iterators advance due to LPU ID changes.

Then, programmers may have redundant LPS transitions in the Computation section of the task that translated as specified will increase overhead. Hence, the compiler tries to reorder stage invocations to fit multiple stage function invocations inside an LPU iteration loop as long as there are no data dependencies precluding such reordering.

Finally, since the computation stages (see Listing 4) are written in a manner oblivious to the data distribution specification which may reorder array indices arbitrary many times (e.g. in Space B of Listing 1), careful consideration is needed to minimize the overhead of transforming an array index to its storage location in the generated C++ stage functions. That HighP5 computation stages use declarative multi-index loops comes in handy in this regard. Consider the ‘saxpy’ computation stage in Listing 4 (from Line 46 to 54). The compiler knows that the row indices are reordered for *u* and *u_block* but column indices remain unaltered. Therefore, it generates a doubly-nested for loop with row traversal in the outer and column in the inner loop to reduce recalculation of data to storage index mapping. This is illustrated in the truncated version of the generated code shown in Listing 6.

```
1 { // scope entrance for parallel loop on index i
2   int i;
3   int iterationStart = uPartDims[0].range.min;
4   int iterationBound = uPartDims[0].range.max;
```

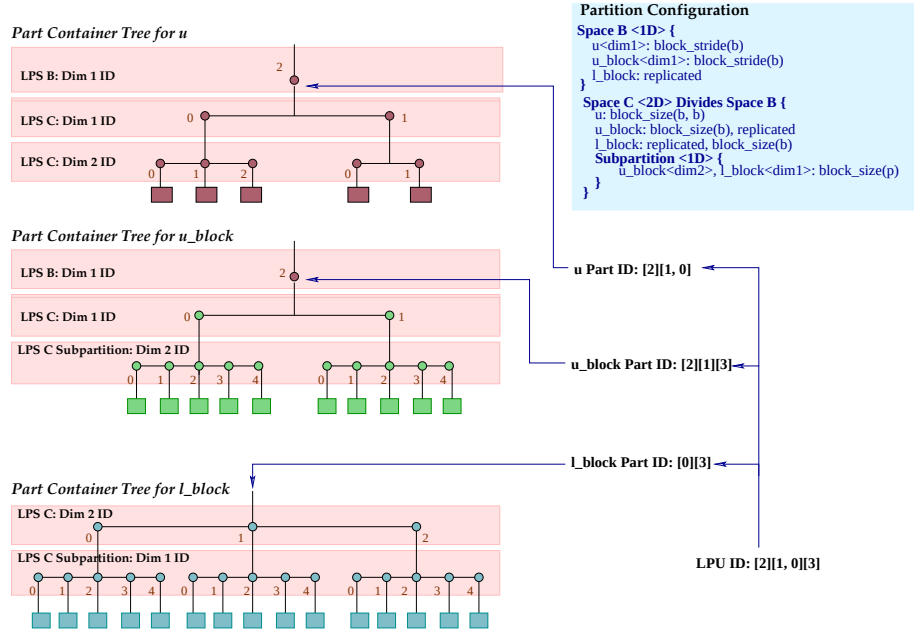


FIGURE 8 | An illustration of the process of retrieving data parts from Data Parts Registry using an LPU ID during block-LU factorization.

```

5 int indexIncrement = 1;
6 int indexMultiplier = 1;
7 // omitted code
8 int iXformed;
9 for (iXformed = iterationStart;
10      indexMultiplier * iXformed <= iterationBound;
11      iXformed += indexIncrement) {
12   // scope start for index retransformation
13   xformIndex = iXformed;
14   partConfig = (&dpu->uPartDims[0])->parent;
15   xformIndex = ((xformIndex / partition.b)
16                * partConfig.count + partConfig.index)
17                * partition.b + xformIndex % partition.b;
18   i = xformIndex;
19 } // scope end for index retransformation
20 long int iu_block0 =
21   ((long) (iXformed - u_blockStoreDims[0].range.min))
22   * ((long) (u_blockStoreDims[1].length));
23 long int iu1 = ((long) (iXformed - uStoreDims[0].range.min))
24   * ((long) (uStoreDims[1].length));
25 // scope entrance for parallel loop on index j
26 int j;
27 int iterationStart = uPartDims[1].range.min;
28 int iterationBound = uPartDims[1].range.max;
29 int indexIncrement = 1;
30 int indexMultiplier = 1;
31 // omitted code
32 for (j = iterationStart;
33      indexMultiplier * j <= iterationBound; j += indexIncrement) {
34   long int jl_block1 = ((long) (j - l_blockStoreDims[1].range.min));
35   long int ju1 = ((long) (j - uStoreDims[1].range.min));
36   total = 0;
37   // scope entrance for parallel loop on index m
38   int m;
39   int iterationStart = u_blockPartDims[1].range.min;
40   int iterationBound = u_blockPartDims[1].range.max;
41   int indexIncrement = 1;
42   int indexMultiplier = 1;
43   // omitted code
44   for (m = iterationStart; indexMultiplier * m <= iterationBound;
45        m += indexIncrement) {
46     long int mu_block1 = (m - u_blockStoreDims[1].range.min);
47     long int ml_block0 = (m - l_blockStoreDims[0].range.min)
48       * (l_blockStoreDims[1].length);
49     total = (total + lpu->u_block[iu_block0 + mu_block1]

```

```

50       * lpu->l_block[ml_block0 + jl_block1]));
51   }
52 } // scope exit for parallel loop on index m
53 lpu->u[iu0 + ju1] = (lpu->u[iu0 + ju1] - total);
54 }
55 } // scope exit for parallel loop on index j
56 }
57 // scope exit for parallel loop on index i
58 return SUCCESS_RUN;

```

Listing 6 A truncated version of the C++ stage function generated by the back-end compiler for the saxpy computation stage of the Block-LU factorization task.

Here, the conversion from data to storage index for u and u_block happens from Line 12 to 18. This strategy of loop ordering based on the data distribution saves computations most of the time. However, when an array index is not a part of a declarative loop, then the index transformation must be done in place, as needed for the index ‘pivot’ in Line 23 of Listing 4.

7 | Synchronization & Communication

In HighP5 programming model, synchronization and communication for shared data within a task only happen at LPS transition boundaries. Consequently, the compiler only focuses on resolving read/write dependencies among computation stages at LPS transitions by inserting proper synchronization/communication code. This is a multi-faceted effort where some of the decisions are made by the compiler front-end, the back-end, and, surprising, even by the RTE initializer. An elaborate approach is implemented in this regard since the optimization of synchronization and, in particular, communication is pivotal for good performance in distributed memory environments.

Note that all LPS transitions may not be explicit in the source code. Hence, the front-end compiler transforms the computation flow specification of the task into a revised intermediate from where all LPS transitions are explicit. This often results in insertions of dummy computation stages in the flow that are only added to expose all read-write data dependencies that are modeled as directional arrows between computation stages. Subsequently, the front-end compiler applies a technique that we call ‘lifting’, to move dependencies between stages inside repetition loops and conditional blocks (e.g., ‘selectPivot’, ‘updateURowsBlock’ of Listing 1) and stages outside them (e.g., ‘copyUpdatedUBlock’) to LPS transition positions that minimize the invocation of dependency resolution routines without violating accuracy. Finally, it investigates all data dependency arrows and removes any redundancies being found. Without the knowledge of the PCubeS description and the mapping of LPSes of the task to PPSes of the hardware, The front-end compiler cannot determine whether a dependency requires synchronization or communication, and how the participants for a dependency resolution should be selected. Therefore, dependency optimization from the front-end compiler ends here.

The back-end compiler compares the PCubeS description of the hardware with the programmer-provided mapping file to determine the *confinement* of a data dependency requirement. A confinement is a branch of the PPS hierarchy rooted at the closest ancestor PPS connecting the source and sink of a data dependency arc (Note that the source and sink can be the same PPS when the related LPSes are mapped to the same PPS or a single LPS used overlapping data partition among its LPUs.). Depending on how LPSes are mapped to the PPSes, confinements may be internal to individual MPI processes or spanning multiple processes. If a confinement is internal to an MPI process and both the source and sync LPSes are using the same data buffer then the data dependency is a synchronization requirement. Otherwise, it is a communication requirement.

7.1 | Resolving Synchronization Dependencies

Synchronization dependencies are resolved by inserting code that does Pthreads barrier synchronizations in appropriate places in the flow execution template. As discussed in Section 3.2, the Pthreads inside an MPI process progress independently as composite PPU controllers and execute parts of the computation flow appropriate for them based on their PPU IDs on LPUs multiplexed to them. A synchronization dependency at any step of their execution means a read-after-write or a write-after-read dependency among two or more PPU controllers at their flow executions’ LPS transition points due to some shared data parts. Proper ordering of read-write is ensured by forcing them to wait on the same Pthreads barrier. Here, the participants of each barrier sync and the number of barrier primitives needed for different interdependent thread groups are deterministically resolved using the hardware’s PCubeS description.

Initialization of all sync primitives is done by the task launcher (Figure 5) before the PPU controller threads are started. Line 20 to 26 of Listing 7 illustrate the technique

of synchronization for update dependency between the ‘updateUColsBlock’ and ‘saxpy’ stages of Listing 1. The idea of lifting is apparent here. Although the stages are in two different repeat loops in the source code, the synchronization is placed outside the loops in the interim C++ code. The synchronization from Line 36 to 39 of Listing 7, on the other hand, is for the update of u inside the ‘saxpy’ stage impacting ‘copyUpdatedUBlock’ of Listing 1, which could not be lifted beyond the Space B LPU iterations without violating accuracy.

```

1  while((lpu = threadState->getNextLpu(Space_B,
2      Space_A, spaceBLpuld)) != NULL) {
3      spaceBLpu = (SpaceB_LPU*) lpu;
4      if (threadState->isValidPpu(Space_B)) {
5          // invoking user computation
6          int stage21Executed = updateucolsblock_stage_20(spaceBLpu,
7              arrayMetadata,
8              taskGlobals,
9              threadLocals, partition, threadState->threadLog);
10         uStage21No4 += stage21Executed;
11     }
12     spaceBLpuld = spaceBLpu->id;
13     spaceBIteration++;
14 }
15 threadState->removeIterationBound(Space_A);
16 // scope exit for iterating LPUs of Space B
17 repeatIteration++;
18 // scope exit for repeat loop
19
20 // resolving synchronization dependencies
21 if (uStage21No4 > 0 && threadState->isValidPpu(Space_B)) {
22     threadSyne->uStage21No4DSyne->signal(repeatIteration);
23     uStage21No4 = 0;
24 } else if (threadState->isValidPpu(Space_C_Sub)) {
25     threadSyne->uStage21No4DSyne->wait(repeatIteration);
26 }
27
28 // scope entrance for iterating LPUs of Space B
29 int spaceBLpuld = INVALID_ID;
30 int spaceBIteration = 0;
31 SpaceB_LPU *spaceBLpu = NULL;
32 LPU *lpu = NULL;
33 while((lpu = threadState->getNextLpu(Space_B,
34     Space_A, spaceBLpuld)) != NULL) {
35     spaceBLpu = (SpaceB_LPU*) lpu;
36     // barriers to ensure all readers have read the last update
37     if (threadState->isValidPpu(Space_C_Sub)) {
38         threadSyne->u_blockStage23No1ReverseSyne->wait();
39     }
40     if (threadState->isValidPpu(Space_B)) {
41         // invoking user computation
42         int stage23Executed = copyupdatedublock_stage_22(spaceBLpu,
43             arrayMetadata,
44             taskGlobals,
45             threadLocals, partition, threadState->threadLog);

```

Listing 7 A part of the thread run function of block-LU factorization interim C++ code illustrating how synchronization is implemented.

The way the synchronization code is generated, a semaphore-based implementation would be possible as an alternative to barrier synchronization that would foster higher independent progress among the relevant PPU controller threads. However, our observation so far in all target platforms is that semaphore and conditional variable based sync primitives scale poorly compared to basic barrier synchronization as the number of participant threads increases. Consequently, both signaling and waiting internally end up waiting on the same barrier primitive. Finally, note the existence of the clause $uStage21No4 > 0$ in the signaling part. As the source for

the dependency may be different PPU controller threads in different iterations, whether a thread actually executed the relevant computation stage is stored in a variable to control the signaling. This is why every stage function returns a status at exit as shown in the last line of Listing 6.

7.2 | Resolving Communication Dependencies

When a data dependency resolution involves communications then – unlike a synchronization requirement – we not only need to know the overall participants but also the individual source-destination pairs and the exact data to be exchanged among each pair. These additional burdens make communication handling far more sophisticated than synchronization.

The only easy scenario is the sharing of update of scalar and fully-replicated data structures. An MPI broadcast among all processes is sufficient. For any array that is partitioned one or more times in the LPS hierarchy, following related questions need to be answered:

1. How does an MPI process determine the other MPI processes it has to communicate with?
2. How does the process know the exact data elements it has to send/receive to/from another process?
3. How to resolve the possible array index ordering discrepancy between sender and receiver side LPSes' data buffers?
4. How to address the issue that the sender and receivers are ready for the data interchange at different time?
5. How to minimize the number of send-receive exchanges per process?

Among the above, only the last question is explicitly performance related. However, there are performance consequences associated with the nature of the solution implemented by the back-end compiler for every other question.

From a high-level perspective, the back-end compiler generates one thread-safe array communicator for each communication dependency in the computation flow. The general communicator architecture is shown in Figure 9.

Inside an array communicator, a communication buffer manager component holds one communicator buffer that collects/distributes all elements from a single receiver/sender MPI process from/to various data parts of the current MPI process. Any reordering of array indices in the data parts get unwinded in the communication buffer if it is for sending and is initiated from the buffer if it is for receiving. The various communication buffers feed data to a single MPI group communication buffer if MPI group communication is feasible for the dependency. Data is directly pulled from the communication buffers for point-to-point MPI communications. As there are, possibly, multiple composite PPU controller threads inside an MPI process; the communication manager has configurable parallel barriers inside it separately for send and receive. As a PPU controller thread reaches the point in the flow execution where it is ready to send data, it waits on the send barrier. When all relevant threads reach the barrier, they collaboratively pull content from data parts to communication buffers then from there to the MPI group buffer (if applicable). Finally, the MPI communication is issued by a

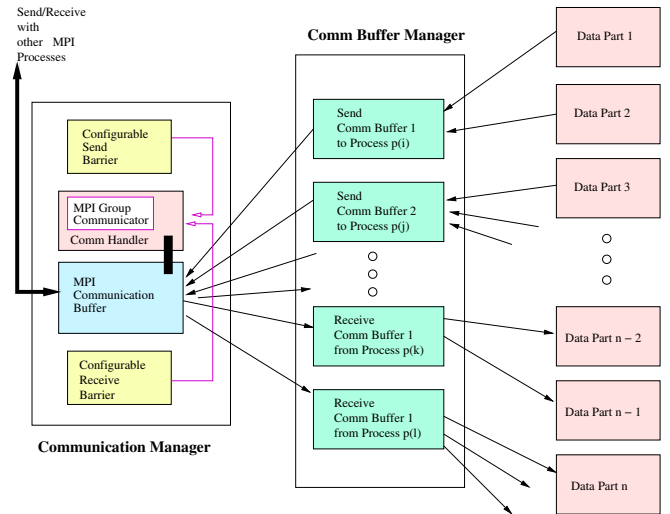


FIGURE 9 | The general architecture of a HighP5 communicator responsible for communicating array parts.

single thread in the group. For a receive, the sequence of activities is just reversed. If a single confinement of communication dependency does not span all MPI processes then separate MPI group communicators are initialized for processes of different confinement groups. The barriers are configurable as they allow skipping and merging of steps (often short circuiting send-and-receive) at task launcher initialization at runtime. Similarly, there are alternative implementations of communication buffers that are ideal for different scenarios that the RTE initializer chooses from. Thus, the back-end compiler generates functions for communicator initialization that takes runtime environment information into account for fine-tuning each communicator part.

Except for an all-to-all broadcast scenario, array communicators always use asynchronous versions of MPI communication primitives to maximize computation/communication overlap between and within sender and receiver processes. The send is issued in the earliest LPS transition after the computation stage invocation that updates the underlying array of interest. Meanwhile, the receive is issued in the latest LPS transition before the first computation stage invocation that requires the update. Line 12 to 30 of Listing 8 shows the invocation of communicator for successive ‘generatePivotColumn’ and ‘updateUColsBlock’ of Listing 1.

```

1 if (threadState->isValidPpu(Space_A)) {
2   // invoking user computation
3   int stage19Executed = generatepivotcolumn_stage_18(spaceALpu,
4     arrayMetadata,
5     taskGlobals,
6     threadLocals,
7     partition,
8     threadState->threadLog);
9   p_columnStage19No1 += stage19Executed;
10 }
11 // communicating updates
12 if (threadState->isValidPpu(Space_A)) {
13   Communicator *communicator
14     = threadState->getCommunicator("p_columnStage19No1");
15   if (communicator != NULL) {
16     if (p_columnStage19No1 > 0) {
17       communicator->send(REQUIRING_COMMUNICATION commCounter5);
18     } else communicator->send(PASSIVE_REPORTING commCounter5);
19   }

```

```

20 p_columnStage19No1 = 0;
21 }
22 // waiting on data reception
23 if (threadState->isValidPpu(Space_B)) {
24 Communicator *communicator
25     = threadState->getCommunicator("p_columnStage19No1");
26 if (communicator != NULL) {
27     communicator->receive(REQUESTING_COMMUNICATION commCounter5);
28 }
29 }
30 commCounter5++;
31 { // scope entrance for iterating LPUs of Space B
32 int spaceBLpuId = INVALID_ID;
33 int spaceBIteration = 0;
34 SpaceB_LPU *spaceBLpu = NULL;
35 LPU *lpu = NULL;
36 while((lpu = threadState->getNextLpu(Space_B,
37     Space_A, spaceBLpuId)) != NULL) {
38     spaceBLpu = (SpaceB_LPU*) lpu;
39     if (threadState->isValidPpu(Space_B)) {
40         // invoking user computation
41         int stage21Executed = updateucolsblock_stage_20(spaceBLpu,
42             arrayMetadata,
43             taskGlobals,
44             threadLocals,
45             partition,
46             threadState->threadLog);
47         uStage21No4 += stage21Executed;
48     }

```

Listing 8 A part of the thread run function of block-LU factorization interim C++ code illustrating use of an array communicator.

If a particular PPU controller thread has no LPU to process for the sender-side stage computation, it still invokes the communicator but with a flag saying that it is ‘PASSIVE_REPORTING.’ This scenario occurs when the number of LPUs are less than the number of PPUs assigned for a particular LPS-PPS mapping. Invoking the communicator is still needed to ensure correct behavior of the send barrier. In both exceptional and regular scenarios a counter is sent as the second parameter to the communicator. This serves two purposes. First, the counter relates asynchronous receives with sends of a particular iteration of communication. Second, if it is found that the same MPI processes are involved in both send and receive then asynchronous receives are issued early during sending data. Then, the later invocation of ‘communicator->receive’ (as in Line 27) can skip several steps of data reception processing.

It is critical to use MPI group communications whenever possible when a single MPI process has data to send/receive to/from many MPI processes. In fact, scalability issues with such communications is so well known [66] that a separate routing network for group communications is common in supercomputers, which MPI group communication primitives take advantage of. A great advantage of portraying the logic of a HighP5 task as a flow of computation across LPSes is that the relative position of the sender and receiver LPSes in the LPS hierarchy and the nature of distribution of the underlying array parts in them dictate what MPI communication primitive is appropriate. Consequently, the back-end compiler generates different types of communicator based on LPS hierarchy and data distribution analysis. The degree of involvement of the positions of LPSes in the choice of a communicator type is illustrated in Figure 10.

The communicator types dictate how data to be exchanged between sender and receiver MPI processes – not what data to exchange. Given a pair of MPI processes i and j , the data

i has to send to j for a dependency from LPS1 to LPS2 can be mathematically described as follows:

$$desc(send_{i,j}) = \left(\bigcup_{part_{LPS1} \in i} desc(part_{LPS1}) \right) \hat{\wedge} \left(\bigcup_{part_{LPS2} \in j} desc(part_{LPS2}) \right) \quad (6)$$

Here, the description of a data part is a multidimensional hierarchical interval function that we discussed in Section 5.1. Given there may be numerous data parts per process, computing intersections on a part-by-part basis is infeasible. To solve this problem, an idea of ‘part-id-tree folding’ is applied on both sender and receiver sides. A folding algorithm recursively generates the most compact interval function description for the data of an LPS in a process of bottom-up contraction of the part-id tree from the interval descriptions of the leaves to the interval description of the root. The approach drastically reduces the number of interval descriptions per process. Subsequently, an interval intersection computation algorithm calculates the interval function for the intersection. These two custom algorithms are part of the common-libraries of RTE that compiler generated code for communication buffers (see Figure 9) invokes during their initiation.

Finally, to determine with what other process to communicate and exactly what data to exchange with a counterpart, an MPI process uses the fact that a PCubeS hierarchy of the execution platform is uniform, and it is possible to get the number of MPI processes using the ‘MPI_Comm_size’ function. Aided with the knowledge of confinement roots of each dependency and how that root LPS is mapped to PPSes of the hardware, an MPI process pretends that it has the MPI rank of each candidate other process within its confinement that it may have to exchange data with. This emulation, during a task initialization, of other processes is done with a memory flag that disables allocation of data buffers but keeps intact part-id-tree constructions. Subsequently, the folding algorithm mentioned above is invoked to find compact interval descriptions of all interesting LPU data of that other process and is used for intersection computations with the current process. Therefore, each MPI process finds out what data it needs to exchange with everyone else for each communication dependency without any overhead communication. However, this strategy consumes time.

The scenarios illustrated in Figure 10 exhaust all possible communication cases to synchronize hierarchically distributed shared/overlapped data expressed as multidimensional periodic interval functions. Noticeably, they cover all communication primitives supported by MPI. This is not surprising because different MPI communication primitives were designed to better support different types of data movement scenarios along hierarchical interconnection networks common in supercomputers and compute clusters. The absence of a communicator resembling MPI reduce/all-reduce is the only exception. In HighP5, a reduce is treated as an explicit language construct (see Line 6 and 11 of Listing 4) that is translated into an appropriate MPI and/or Pthreads reduce implementation outside the aforementioned communicator framework. The reason for treating reduce differently is that

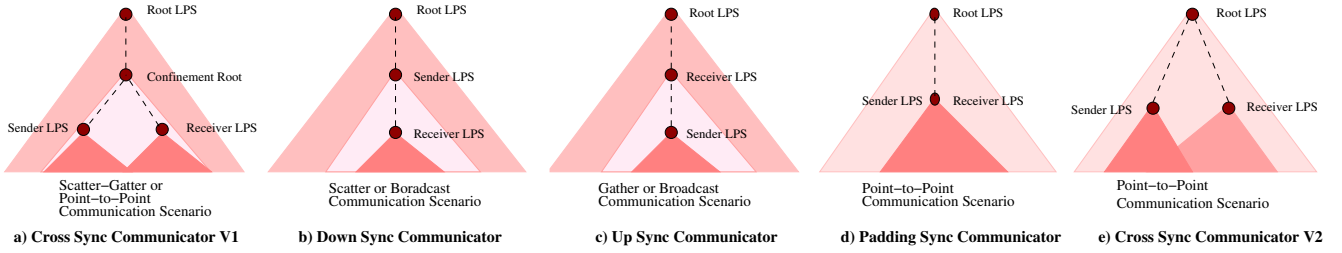


FIGURE 10 | An illustration of how relative positions of the LPSes in the LPS hierarchy involved in a data dependency determine the nature of MPI communication primitives.

a reduce represents a data conversion (sum, avg, max, etc.) that produces a new result instead of synchronizing updates on shared data addressed by the communicators.

8 | Environment Management

Programmers launch the various tasks of a multi-task HighP5 program from a coordinator module/function (called the ‘Program’ in HighP5 terminology). As we stated earlier, the tasks are related to each other by their environment dependencies. To illustrate this concept, we deviate from our running example of block-LU factorization program and present the coordinator module of a multi-task HighP5 conjugate gradient program in Listing 9. Here the solution of a sparse system of equations is computed using iterative interactions among a vector dot-product, vector addition, and sparse matrix-vector multiplication tasks (the coordinator module’s logic for this program is easy to understand from the comments embedded in the code). The back-end compiler generates the same intermediate code (equivalent of a main function) for all MPI processes with the decision-making logic embedded in places for the processes to participate in or skip a particular task launch based on their MPI rank and task mapping configuration.

```
1 Program (args) {
2   // creating environment objects for component tasks
3   vaEnv1 = new TaskEnvironment(name: "Vector Add")
4   vaEnv2 = new TaskEnvironment(name: "Vector Add")
5   dpEnv = new TaskEnvironment(name: "Vector Dot Prod")
6   nmEnv1 = new TaskEnvironment(name: "CSR Matrix Vector Mult")
7   nmEnv2 = new TaskEnvironment(name: "CSR Matrix Vector Mult")
8
9   // make the argument sparse matrix stored in compressed row
10  // format from files to be read during first-time execution
11  // of the matrix-vector multiply task
12  bind_input(nmEnv1, "columns", args.arg_matrix_cols)
13  bind_input(nmEnv1, "rows", args.arg_matrix_rows)
14  bind_input(nmEnv1, "values", args.arg_matrix_values)
15
16  // bind the prediction (x_0) and known vector (b) to the tasks'
17  // environment that use them initially
18  bind_input(vaEnv1, "u", args.known_vector)
19  bind_input(nmEnv1, "v", args.prediction_vector)
20
21  // run the conjugate gradient logic
22  iteration = 0
23  maxIterations = args.maxIterations
24  do {
25    // calculate A * x_i
26    execute(task: "CSR Matrix Vector Mult";
27            environment: nmEnv1; partition: args.r)
28    // determine the current residual error r_i = b - A * x_i
29    vaEnv1.alpha = 1
```

```
30    vaEnv1.v = nmEnv1.w
31    vaEnv1.beta = -1
32    execute(task: "Vector Add";
33            environment: vaEnv1; partition: args.b)
34    // determine r_i.r_i as the residual norm
35    dpEnv.u = dpEnv.v = vaEnv1.w
36    execute(task: "Vector Dot Prod";
37            environment: dpEnv; partition: args.b)
38    norm = dpEnv.product
39    // in the first iteration setup duplicate environmental
40    // references for the sparse matrix components
41    if (iteration == 0) {
42      nmEnv2.columns = nmEnv1.columns
43      nmEnv2.rows = nmEnv1.rows
44      nmEnv2.values = nmEnv1.values
45    }
46    // determine A * r_i
47    nmEnv2.v = vaEnv1.w
48    execute(task: "CSR Matrix Vector Mult";
49            environment: nmEnv2; partition: args.r)
50    // determine dot product of r_i to A * r_i
51    dpEnv.v = nmEnv2.w
52    execute(task: "Vector Dot Prod";
53            environment: dpEnv; partition: args.b)
54    // update the step size to (r_i.r_i) / (r_i.(A * r_i))
55    alpha_i = norm / dpEnv.product
56    // calculate the next estimate x_i = x_i + alpha_i * r_i
57    vaEnv2.u = nmEnv1.v
58    vaEnv2.alpha = 1
59    vaEnv2.v = vaEnv1.w
60    vaEnv2.beta = alpha_i
61    execute(task: "Vector Add";
62            environment: vaEnv2; partition: args.b)
63    // prepare x_i for the next iteration
64    nmEnv1.v = vaEnv2.w
65    iteration = iteration + 1
66  } while iteration < maxIterations and norm > args.precision
67
68  // store the final solution vector in an output file
69  bind_output(vaEnv2, "w", args.solution_vector)
70 }
```

Listing 9 The program coordinator module for a multi-task implementation of the conjugate gradient program.

To implement the concept of successive task executions modifying portions of the environment and there is no accumulation of the results during tasks transitions, the front-end compiler converts the program coordinator module into a flow specification just like the Computation section of a task, then it does a cross-task dependency analysis to setup the data dependency arcs among them. For the back-end compiler, every inter-task data dependency is a potential communication requirement. However, a communicator setup to resolve a data dependency between two tasks is more involved than doing so for an LPS transition within a single task. The reasons for the additional complexity are two. First, it is difficult to compare the data distributions of two tasks using

the same data since their LPS hierarchies and LPS-PPS mapping configurations are unrelated. Second, several other task launches may happen between the data updater task and the later task requiring the update, which delays the dependency processing.

The strategy in this regard is to move all task-global data structures under the supervision of environment manager component of the RTE (see Figure 4). External input-outputs (e.g., Lines 12-14 and Line 69 of Listing 9), updates sharing between tasks (e.g., Line 30 and 35), and data generation within a task (e.g., ‘mvmEnv1.w’ of Line 30) for such data structures are all registered as environment instructions in the intermediate C++ code. The back-end compiler generates garbage collection instructions for any task-global data structures that are not needed by any subsequent task executions or external output. These instructions are executed by the task executor during a task initiation and termination based on the instruction type. Consequently, only useful data remain in the collective memory of the MPI processes and unnecessary data are cleaned up. This is even true for multiple versions of the same data within a single task which are generated when the task uses different index-reordering distributions for the data in various LPSes. In that case, the buffers corresponding to the stale versions are reclaimed, keeping the last updated one intact. In addition to the buffers, the interval configuration corresponding to the hierarchical distribution is also retained. The environment manager keeps track of each such data structure using a unique environment linking key.

When a later task needs a data structure updated by an earlier task, it searches the environment manager with the same environment linking key against a data transfer type environment instruction where it supplies its data distribution requirement. The environment manager then compares the stored version’s distribution with the distribution asked for and does MPI scatter-gather communications to re-distribute the data, if needed. If the two distributions are equivalent, then no communication occurs and the stored buffer is shared with the new task. If the comparison result dictates that an internal reorganization is sufficient, then again no communication is involved, rather a new version is created for the data within relevant MPI processes.

The current strategy for cross-task data dependency resolution works. However, there are improvement opportunities, such as, avoiding some recurrent computations when the updater and dependent tasks occur within loops and doing the dependency resolution asynchronously and early enough to reduce parallelism opportunity loss during task transitions. We need more sophisticated static analysis in the front-end compiler and rethink the architecture of environment manager and task executors to pursue those improvements.

9 | Experiments and Analysis

To evaluate the efficiency of HighP5 distributed memory compiler, we compared the performance of HighP5 executable with optimized handwritten MPI implementations for three building block applications on the Fugaku supercomputer [67][68]. We had a ‘small project’ allocation in the machine for

both compiler improvement and performance testing, and we were able to scale our experiments up to 100 compute nodes. Each Fugaku compute node has 48-core Fujitsu A64FX processor arranged in a group of four 12-core NUMA nodes with 8 GB HBM2 integrated memory in them, which gives a total 32 GB memory per compute nodes. The compute nodes are connected with the Tofu-D interconnect, which is a 6-D torus network with injection bandwidth of 40.8 GB/s per node. In general, we had a very fast environment in terms of both computation and communication capacities. Fugaku login nodes are, on the other hand, Intel machines. Hence we used the cross-architecture ‘mpiFCCpx’ provided in the environment as the final C++ compiler to generate executable for the Arm architecture from the intel machines. O3 optimization flag was enabled in the cross platform MPI compiler during compilation for both the reference and HighP5 codes. In particular, we were interested in assessing how HighP5 programs fare against handwritten MPI programs for applications involving heavy communications. Consequently, the three building block problems of Table 1 were chosen that have a good balance between their computation and communication complexities. In addition, the nature of communication dependencies is also quite different in these problem solutions. Block LU factorization requires directional broadcasts (MPI Bcast, Gather, Scatter, and MPI Reduce) communications in each iteration, finite value estimation using Jacobi iterations (which is commonly called *5-point stencil*) involves pair-to-pair communication (MPI Send/Receive) per iteration for overlapping data synchronization, and conjugate gradient requires multiple all-to-all broadcasts (MPI Allgather and scatter) for each gradient descent step.

As the reference implementations are pure MPI codes, we used multiple MPI processes per node when running them. Although Fugaku compute nodes have 48 cores, we found that the reference MPI implementations’ performance do not improve – rather, they slightly degrade – if more than 32 cores per node are used. Hence, block LU factorization and 5-point stencil reference programs were run with 32 MPI processes per node configuration. However, for the conjugate gradient program, the number of processes per node was further reduced to eight. This is because the Fugaku environment has a limit of 30 GB memory per node for programs and the input sparse matrix of the conjugate gradient is replicated in all processes, causing a memory fault for large inputs if we increased the process count beyond that. Therefore, the PCubeS model of the platform for HighP5 compiler is also adjusted to have the same degrees of parallelism per-node to maintain parity with the reference codes. However, the HighP5 compiler generates MPI+Pthreads hybrid program as discussed in the earlier section where there is one process per node and intra-node parallelism is achieved using Pthreads (we made an exception to this nature by changing the PCubeS model of the platform during conjugate gradient program compilation for a reason that we will explain later). Furthermore, the operating system installed on the compute nodes of Fugaku does not expose the physical core to logical core ID mapping in the `/proc/cpuinfo` file. Neither does it provide information regarding which core groups share what L2 cache segment. These are information that the HighP5

TABLE 1 | Descriptions of the Building-block Applications being Used for Performance Experiments

	Description	Time Complexity	Communication Complexity
Block LU Factorization	Triangulation of a $n \times n$ dense matrix	$\mathcal{O}(n^3)$	$\mathcal{O}(n^2)$
Finite Value Estimation	Heat propagation estimation on a $n \times m$ plate for o iterations using p processors	$\mathcal{O}(nmo)$	$\mathcal{O}(mpo)$
Conjugate Gradient	Solving a sparse system of n equations using o gradient descent iterations	$\mathcal{O}(n^2o)$	$\mathcal{O}(no)$

compiler uses for careful pinning of Pthreads to cores that maximizes cache sharing and reduces synchronization costs. Therefore, the HighP5 versions were slightly at disadvantage that could not be overcome. The reference and HighP5 codes for each node configuration were run 10 times. The standard deviations in 10 runs for all programs are insignificant compared to the average execution times.

Figure 11 compares the average execution time of a block-LU factorization HighP5 program with that of the reference MPI implementation as we step-by-step double the number of nodes from 1 to 64 for a large input matrix. We observe that as we increase the number of nodes, HighP5 program's performance keeps improving at a higher rate than the reference and eventually the former surpasses the latter. The reason for better scalability in HighP5 program is that its communication overhead grows at a rate significantly slower than that of the reference implementation since the former uses 32-times fewer processes than the latter. The question is why performance is lagging for small number of nodes. Our finding is that thread synchronization cost is very high in Fugaku compute nodes when the number of threads participating in a synchronization is quite high, which is the case for several intra-node data dependencies in the block-LU program. When the growth of communication overhead outweighs the constant slow synchronization cost inside individual nodes, the HighP5 program surpasses the reference in performance.

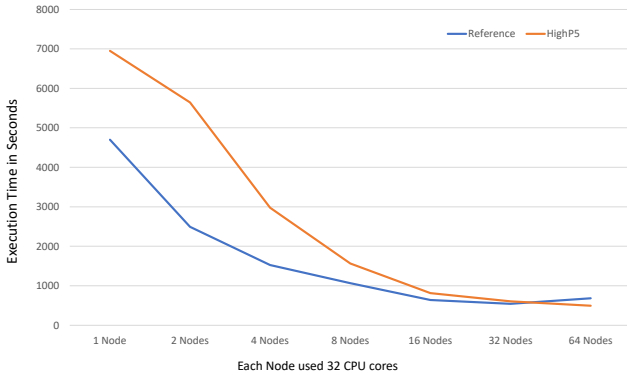
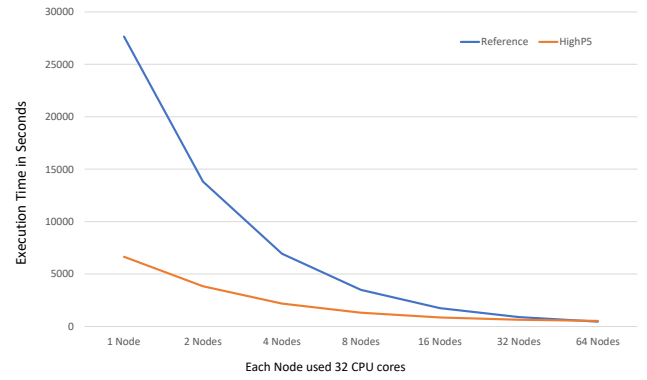
**FIGURE 11** | Execution time comparison for block LU factorization of a 30720^2 Square Matrix for different number of nodes.

Figure 12 demonstrates a reversed trajectory for the average execution times of HighP5 and reference implementations for the 5-point stencil problem. Here, the reference implementation initially lags behind and eventually catches up with the HighP5 implementation. Surprisingly, the reason for this behavior is again the nature of communication overhead for this program. The stencil algorithm has point-to-point data sharing among processes/threads handling neighboring overlapping partitions. As the number of nodes increases, the

computation/communication ratio per thread keeps decreasing, keeping the communication overhead almost fixed due to its point-to-point nature. Consequently, the advantage HighP5 implementation has due to doing more computation per-process keeps decreasing for the fixed input size and eventually both implementations become roughly equivalent. This indicates that for a larger input size, HighP5 implementation should again perform better than the reference for 64 nodes and beyond.

**FIGURE 12** | Execution time comparison for finite difference calculation on a 204800 by 2048 rectangular plate using hundred thousand Jacobi iterations.

An important finding from the first two scalability comparison experiments is that a major obstacle to further performance improvement in HighP5 program is the slowness of group synchronization on the execution platform (in fact, the operating system of the compute nodes). This is a particular concern for our last problem, conjugate gradient, which has three all-thread synchronizations as part of three all-to-all communication per gradient descent iterations. Therefore, we adjusted the PCubeS model of the platform so that the HighP5 compilers generate pure MPI codes for this problem (this adjustment requires a simple property update in the PCubeS model of the hardware). We observe that with this change, the average execution times of the HighP5 implementation are hardly discernible from that of the reference implementation for the conjugate gradient problem for all degrees of parallelism.

From the discussions of the previous sections, it is apparent that the intermediate C++ code that the back-end part of the HighP5 compiler emits is significantly more complicated than the reference implementations of the three problems that have problem-specific optimizations related to MPI communication and buffer management. To better understand how different aspects of the generated code affect the overall running time of the executable, we ran HighP5 and reference implementations using 100 nodes and time-logging enabled

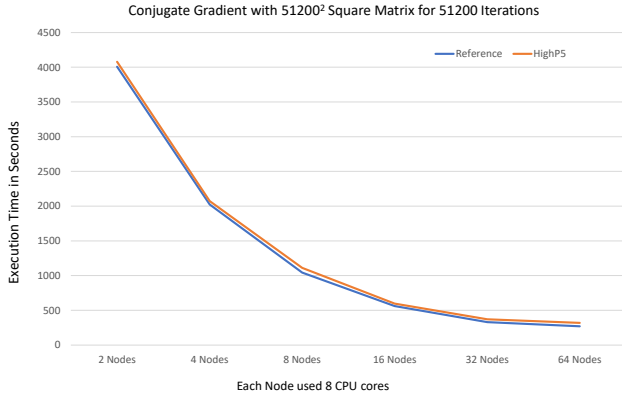


FIGURE 13 | Execution time comparison for conjugate gradient computation for a 95% sparse 51200^2 square matrix for 51200 iterations.

for larger input sizes for the three building block problems. The input size is chosen in a manner to ensure that each program takes a minimum of 5 minutes to complete so that meaningful measurements can be taken. Figure 14 compares the HighP5 version's decomposition of average execution time with that of the reference where the execution times of reference implementations have been normalized to 1. In the charts, memory management time refers to the time taken by input/output and all other data structures setup, and communication time includes both the intra-node synchronization and inter/intra-node data movement cost.

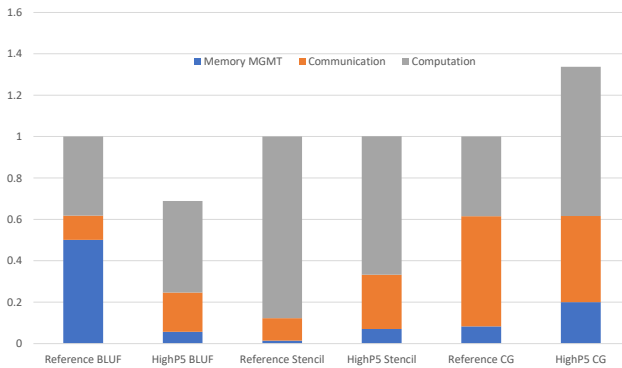


FIGURE 14 | Detail breakdown of the normalized execution times for three building block programs with large input sizes and 100 nodes scenario.

We observe that HighP5 block-LU factorization is faster than the reference implementation for the 100 nodes scenario, and the main contributor to the worsening performance in the reference is the memory management overhead. We found that for 100 nodes and 32 MPI processes per node in the reference implementation, concurrent file reading for the input matrix is extremely slow. The HighP5 implementation, on the other hand, uses only 100 open file handles. Setting up other resources (e.g., communicators, and part-id-trees) is actually costlier in HighP5 than in the reference implementation

which only needs some buffer allocations. The communication cost is slightly higher in the HighP5 implementation. There are two reasons behind this. The first reason is the slow intra-node synchronization that we discussed earlier. Second, the reference implementation avoids two broadcast communications for scalar variables by replicating some calculations in each MPI processes. Taking advantage of this kind of optimization opportunities in a generic way requires a sophisticated static analysis in the compiler. Finally, the additional computations due to reordered index transformation is almost unnoticeable in the HighP5 implementation's computation time.

The reference and HighP5 implementations are still on par for the 100 node scenario for the 5-point stencil program. However, the decomposition of execution times is quite different. The HighP5 implementation again takes more time for memory management compared to the reference due to additional data structure setups. For this problem, I/O operation scaled well for a large number of MPI processes due to the fact that they sequentially read data from consecutive chunks of the file, which was not the case for block-LU factorization. The communication cost in HighP5 implementation is higher than that of the reference, again, due to slow synchronizations. However, a smaller cost of computation offsets the higher communication cost. A sensible explanation for this is better cache utilization through cross-thread collaboration in the HighP5 implementation that the reference cannot avail. The HighP5 implementation for the conjugate gradient is slightly slower than the reference implementation. One reason is the higher memory management overhead, which was expected. However, the other reason, a higher computation cost for this problem is a bit surprising. We suspect that since computation per thread/node in this problem is significantly lower than that of the other problems, the overhead computation has a more dominant impact here. We should attempt further improvements in the code generation strategy for computation stages of HighP5 tasks. Since both reference and HighP5 implementations are pure MPI codes for this problem, no slowness in communication time is observed for this problem. However, we are unclear why the communication is slightly faster in the HighP5 version.

9.1 | Impact of Compiler Optimizations

To understand the effectiveness of various compilation techniques for improving an executable's running time, we ran different versions of the generated executables with the same input and parameter configurations for the 64-node cases discussed earlier and compared their average normalized execution time with the latest version's performance as the reference baseline. In this experiment, each successive version cancels a major compiler optimization present in the previous version by backtracking on our compiler's subversion repository. Compiler optimizations canceled in this manner, in their order of removal, are as follows.

1. **Version 1 (Disable Caching):** cancels careful utilization of caching during filtration, ordering, and grouping of communication buffers inside communicators from the latest version.

2. **Version 2** (Disable Buffer Processing Optimization): removes utilization of thread pools and optimization on the number of steps involved in communication and synchronization operation from Version 1.
3. **Version 3** (Disable Data Transfer Optimization): disables the exploitation of blocked transfer opportunities between data parts and communication buffers during communications from Version 2.
4. **Version 4** (Disable Parallel Initialization): turns off parallelization in data structure setup and input processing during task launching from Version 3.
5. **Version 5** (Disable Sync Primitive Optimization): reverts back to a less sophisticated implementation of synchronization using more semaphore and condition variables from Version 4.

Figure 15 illustrates how the performance of block LU factorization deteriorates at each step compared to the performance of the HighP5 baseline version. We observe the most notice-

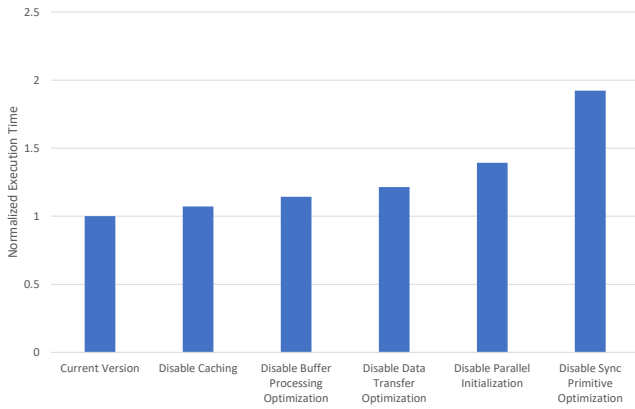


FIGURE 15 | Normalized execution times of different HighP5 block-LUF executable versions for the 64 node scenario.

able increase in execution time in the last step when the synchronization primitive’s optimization is canceled. In block LU factorization, most data movements and synchronizations occur up and down on a single LPS hierarchy. That is, often many PPU controller threads are waiting for update from a single local/remote controller or vice versa. As a result, inefficiencies due to extra wait/signaling in semaphores hurt performance the most. However, the findings of Figure 15 show that all compiler optimization techniques have a noticeable impact on the executable’s performance in this problem.

The performance of 5-point stencil follows a different trajectory, as shown in Figure 16. We observe the most severe impact on the running time after disabling data transfer optimization between data parts and communication buffers. 5-point stencil involves periodic point-to-point exchanges for overlapped multi-versioned data parts among PPU-controllers in both intra and inter-process scenarios. Optimization of this operation is critical for overall program performance. Compared to block LU factorization, the last synchronization primitive optimization has only a marginal impact. This is again due to a lack of group synchronizations in this problem. Here, a composite PPU controller only

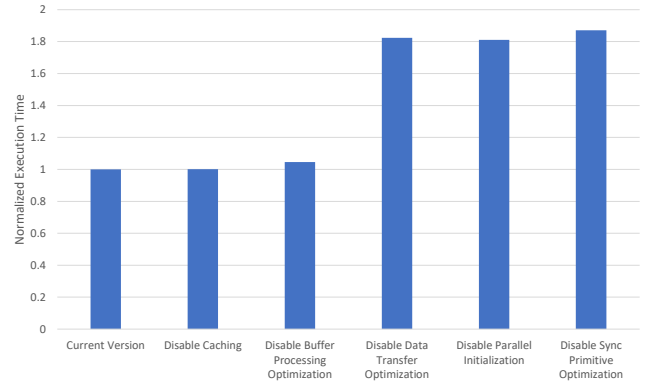


FIGURE 16 | Normalized execution times of different HighP5 iterative stencil executable versions for the 64 node scenario.

synchronizes with the immediate previous and next PPU controller.

Another noticeable aspect is the apparent ineffectiveness of parallelization of initial data structure setup and input processing during task launches in this problem. For large input sizes, initialization time is dominated by file reading operations. The difference in the decrease of performance in block LU factorization and 5-point stencil when parallelization is removed is most likely due to native OS’s file-system optimization for sequential file access effective in 5-point stencil but not for striped file reading that takes place in block LU factorization. If we refer to the execution time decomposition of Figure 14, a significantly higher initialization time for the reference implementations of block LU factorization compared to that of 5-point stencil also supports this assessment.

The performance of the HighP5 conjugate gradient program as we replel various compiler optimizations exhibits yet another surprising trajectory, as depicted in Figure 17.

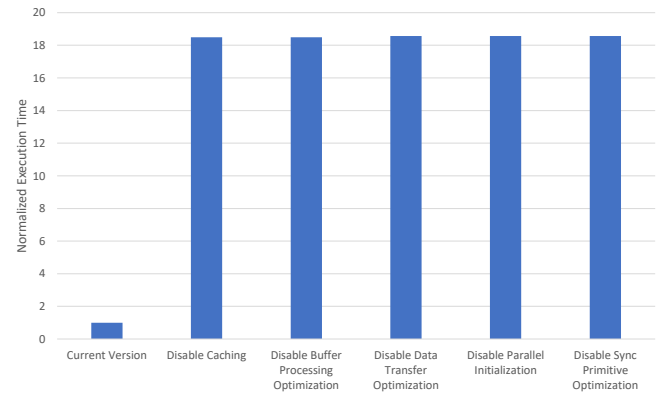


FIGURE 17 | Normalized execution times of different HighP5 conjugate gradient executable versions for the 64 node scenario.

The disabled caching in the filtration, ordering and grouping of communication buffers worsens the running time by more

than 18 times. This happens because conjugate gradient involves communication between LPS hierarchies. In the 64 node and 8 processes per-node scenario, each PPU controller has to communicate with 511 other PPU controllers, where each communication involves at least one buffer management. At that range, the cost of buffer processing is severely felt in the overall running time if caching is not applied. Subsequent optimizations involving parallelization are ineffective in this problem, due to the pure MPI (a separate process per-core) configuration we used to have parity with the reference baseline discussed earlier.

Overall, the lesson we learn from this experiment is that there is no particular optimization that is the most important. Similarly, there is no priority ordering among the different optimizations. Rather, the compiler should not miss any optimization opportunities to generate executables competitive with optimized low-level implementations.

10 | Conclusion

In this article, we discussed the overall architecture and various aspects of executable generation of the HighP5 programming language compiler for supercomputers and compute clusters that connect multicore/multiprocessor CPU nodes through fast interconnection networks, commonly known as distributed memory machines. As HighP5 uses a unified ‘PCubeS’ architecture abstraction to expose salient features of distributed memory machines, instead of letting programmers directly reason about those features, the compiler has significant additional responsibilities regarding generating efficient binaries from source codes that are not concerns for low-level programming paradigms. The article discusses these responsibilities and their complexities in detail and explains how the HighP5 compiler addresses them. The experimental results on some frequently used building block problems suggest that the compiler is largely successful in generating binaries that are competitive with efficient handwritten, low-level alternatives. However, there are opportunities and challenges associated with further improvements. The explanations and analyses presented in the article should help practitioners and enthusiasts of parallel computing languages and tools development.

As the next step of our research as compiler and language designers, we want to enhance HighP5 programming language with the capabilities of dynamic parallelism that is needed to support problems that occur in fluid-dynamics and astrophysics. Then we will look at the scope of improving the distributed memory compiler further to ensure that updated strategies for executable generation benefit most important programming patterns. **In particular, our experience with the conjugate gradient problem in Fugaku supercomputer shows that deciding when to create a thread and when to create a process just based on the architectural breakdown of the execution platform into intra-node and inter-node regions is insufficient. Rather, operating system’s implementation of synchronization primitives and the intra-node span and frequency of synchronizations in a HighP5 program should be taken into account for ideal executable generation. We believe, benchmarking performance of standard building block**

programs (e.g., Berkeley dwarf applications [69]) on the target execution platform then using their performance to guide further static analyses about executable’s nature selection in the HighP5 compiler front-end is the right direction for improvement in this regard.

There is another HighP5 compiler at its primitive stage that generates binaries for hybrid environments that have both GPUs and CPUs in supercomputer/cluster nodes. However, optimizing the distributed memory compiler to the extent possible is more urgent as the compiler for hybrid environments needs to answer the same questions regarding CPU host usage and cross-node communication optimization. It just has additional responsibilities regarding efficient utilization of the GPUs.

Author Contributions

Muhammad Nur Yanhaona is the primary contributor of this research work. He worked under the supervision of Andrew Grimshaw.

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Financial Disclosure

None reported.

Conflicts of Interest

The authors declare no conflicts of interest.

Supporting Information

The source code of the compiler, HighP5 and reference source codes used for experiments, and the detail of performance results reported in this paper are all available in the Fugaku branch of the Github repository [HighP5-Compilers](#).

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